

Technological Change and the Cost of Redistribution

John Sturm Becko^{*}

Princeton

André Sztutman[†]

Carnegie Mellon

Anchi Xia[‡]

MIT

November 30, 2024

Abstract

How does technological change affect the fiscal cost of redistributing income through taxes? We address this question in light of the well-documented heterogeneity in elasticities of labor supply and taxable income. When the same households who are responsive to wage changes are also responsive to tax changes, a uniform increase in productivity disproportionately shifts more tax-elastic types to higher incomes. As a result, it raises the cost of redistributing income away from the top. We provide novel evidence showing that this sorting mechanism is quantitatively powerful and, in recent decades, outweighs a countervailing mechanism resulting from higher income inequality.

^{*}Julis Rabinowitz Center, Princeton University. Email: johnsbecko@gmail.com.

[†]Tepper School of Business, Carnegie Mellon University. Email: andresztutman@gmail.com.

[‡]Massachusetts Institute of Technology. Email: bryantxia2435@gmail.com.

1 Introduction

It is an often repeated claim that US income inequality has increased in recent decades [Piketty and Saez, 2003, Piketty et al., 2018, Auten and Splinter, 2024]. This observation suggests that taxes may have failed to “keep up” with increasing inequality and that redistributive considerations may have lost their sway in shaping US tax policy. However, this discourse has—in focusing on normative preferences to redistribute income—largely overlooked the possibility that the positive costs of increasing marginal taxes may have changed over the same period.

In this paper, we shed new light on secular changes in inequality and income taxation, focusing on the fiscal costs of redistributing income through the income tax system. Importantly, we address this topic in light of the well-documented and pervasive heterogeneity in taxable income elasticities, even within income levels [Eissa and Liebman, 1996, Gruber and Saez, 2002, Becko and Sztutman, 2024]. We show that within-income elasticity heterogeneity generates a powerful sorting mechanism that drives up average elasticities at high incomes whenever productivities increase. This prediction is in line with suggestive evidence that, average elasticities of taxable income at the top have indeed risen in recent decades [Rubolino and Waldenström, 2019]. Quantitatively, we argue this effect has been the driving force behind recent changes in the costs of redistributing income, overshadowing a countervailing effect due to increased income inequality.

Our starting point for this argument is a new formalization of what it means for redistribution to be costly. We define the cost of redistribution in terms of redistributive tax reforms that reduce taxes by one dollar below an income percentile p and increase them by $p/(1-p)$ dollars above p . Although such reforms are budget-neutral if households do not adjust their behavior, they in general affect tax revenues by distorting labor supply decisions. The cost of redistribution at percentile p is the size of the lump-sum tax required to make up these lost revenues and thereby finance the redistributive reform. We show that this definition is closely related to the slope of the Pareto frontier and familiar statistics that measure households’ taxable income elasticities and the shape of the income distribution.

How does the cost of redistribution change in response to technological changes, such as those that may have shaped the US income distribution in recent decades? Our analysis provides a decomposition that—in a simple Mirrleesian environment—highlights two key economic forces. On one hand, technological changes can impact the shape of the income distribution if either they have heterogeneous wage effects or households at different incomes respond differently to wage increases. On the other hand, technological changes can impact the average elasticity of taxable income at each income level if households at the same income respond differently to wage increases.

A simple *stretching* mechanism underlies changes in the shape of the income dis-

tribution and their effects on the costs of redistributing income. If productivity gains are proportionally larger for higher income levels, then the income distribution stretches out as productivity increases. The same is true if productivity gains are uniform but elasticities are increasing in income. This affects the fiscal impacts of tax reforms that increase marginal taxes around some income level z . When the income distribution is more spread out, such reforms change fewer households' marginal taxes relative to the number of inframarginal households for which they increase the level of taxes. This is relevant because (absent income effects) marginal households determine the reform's distortions while inframarginal households determine how large of transfers it makes. Consequently, a distribution-stretching technological change reduces the costs of redistribution. This effect is precisely captured by the change in the first-order local Pareto shape parameter of the income distribution, i.e. changes in $-d\log(1 - F(z))/d\log z$, where $F(z)$ is the cumulative distribution function of income z . In the last three decades, increased inequality has been responsible for an decrease of 15% in the shape parameter of the income distribution at the top, which implies that the *stretching* of the income distribution has contributed to a 15% reduction in the cost of redistributing income away from top earners.

An equally simple *sorting* mechanism relates changes in productivities and heterogeneity in elasticities to changes in average elasticities at each income level and, through them, the costs of redistributing income. If elasticities are heterogeneous within an income level, then as wages increase, more elastic types will increase their incomes more. At incomes where the slope of the income density is sufficiently negative (as it is at high incomes), these high-elasticity types will become relatively more prevalent relative to low-elasticity types. Therefore, the average elasticity at that income level will increase. Conversely, if the density of the income distribution is increasing, as it is at low incomes, then the average elasticity will decrease.

The crucial assumption underpinning this sorting mechanism is that the households who respond the most to productivity changes are also the households who respond the most to changes in marginal tax rates. However, there is no readily available evidence on the correlation between income elasticities with respect to tax changes and income elasticities with respect to wage changes. We fill this gap using data from the Survey of Income and Program Participation and machine learning methods that allow us to estimate heterogeneous wage and tax elasticities [Athey et al. \[2019\]](#). More concretely, we estimate heterogeneous elasticities of income with respect to marginal tax changes by combining instrumental forests and a [Gruber and Saez \[2002\]](#) identification strategy: we use simulated tax changes at initial incomes as an instrument for actual changes in marginal taxes in the instrumental forest while controlling for marital status, state-time fixed effects and controlling non-parametrically for initial incomes. We estimate heterogeneous elasticities with respect to hourly wage changes using a similar method

and instrumenting for hourly wage changes with a leave-one-out measure of changes in the worker’s initial industry. We find that these two elasticities are strongly correlated (≈ 0.7). At the top of the income distribution we estimate an income-conditional correlation of around 0.6, with confidence intervals that do not rule out a perfect correlation between the two elasticities but do rule out a correlation lower than 0.3.

We then combine our theoretical results and empirical estimates to quantify changes in the cost of redistribution over the last 30 years in the US. On one hand, the *stretching* mechanism has contributed to decreasing costs the costs of redistributing income from the top of the income distribution. We infer the strength of the stretching mechanism from the change in the Pareto shape of the income distribution. In 1994, it was estimated to be around 1.7, whereas in 2023, it declined to around 1.5. That amounts to approximately a 15% decrease in the costs of redistribution at the top of the income distribution.

On the other hand, the strength of the *sorting* mechanism has contributed to increasing the costs of redistributing income. Combining estimates of variances of elasticities, the correlation between elasticities out of wage changes and out changes in marginal rates, with the local shape of the income distribution, we can infer that the sorting mechanism has contributed to approximately a 45% increase in the cost of redistributing income from the top.

On net, we estimate that the cost of redistribution away from top US earners has risen by 30% since 1994. While we intentionally focus on changes in redistribution’s positive costs, our findings implicitly identify changes in policy judgments about redistribution’s normative benefits [Bourguignon and Spadaro, 2012]. The constancy of marginal taxes on top earners over the same period, therefore, is consistent with an *increased* social value on redistribution in the eyes of policymakers. This casts doubt on the popular narrative that taxes have “failed to keep up” with rising income inequality.

Related Literature First, this paper contributes to a long literature on the effects of technological change on income inequality [Katz and Murphy, 1992, Piketty and Saez, 2003, Autor et al., 2003, Acemoglu and Restrepo, 2020, Acemoglu et al., 2022], and the literature on the optimal tax response to technological changes [Ales et al., 2015, Scheuer and Werning, 2017, Heathcote et al., 2017, Wu, 2021]. Relative to this literature, we present a novel decomposition of the effects of general technological change that ultimately impact salaries on the costs of redistributing income.

Our decomposition isolates the cost side of the trade-off governments face when they think of redistributing income. By inferring trade-offs from the observed choices, this approach also builds on the inverse optimum literature [Christiansen and Jansen, 1978, Bourguignon and Spadaro, 2012]. Moreover, it builds on a long literature that has proposed a variety of measures to evaluate different policies, such as the marginal value of public funds [Hendren, 2020, Hendren and Sprung-Keyser, 2020].

Our results build on the literature on non-linear income taxation with multi-dimensional heterogeneity [Scheuer and Werning, 2016, Jacquet and Lehmann, 2021, Bierbrauer et al., 2023, Becko and Sztutman, 2024], by highlighting the interaction of technological changes and productivity increases with heterogeneity in elasticities of taxable income in a standard Mirrlesian model of nonlinear taxation.

We also contribute to the literature on tax avoidance and optimal taxation [Feldstein, 1999, Slemrod, 1995, 2002, Chetty, 2009], as well as the empirical literature on heterogeneity in labor supply responses and elasticities of taxable income [Eissa and Liebman, 1996, Blau and Kahn, 2007, Gruber and Saez, 2002, Kumar and Liang, 2020, Becko and Sztutman, 2024]. We estimate of elasticities of taxable income with respect to marginal tax changes and elasticities of taxable income with respect to wage increases, and document a high correlation between these two concepts. This implies that productivity increases are likely to be a driving force behind the reallocation of different elasticity types across the income distribution and a key driver of changes in the costs of redistributing income.

2 Model

We study a static Mirrlesian economy, allowing for heterogeneity in household elasticities of labor supply and taxable income.

2.1 Environment

The economy contains a unit measure μ of households $h \in \mathcal{H}$, a representative firm, and a government that levies taxes and makes expenditures. The economy contains a single consumption good and one factor per household.

The representative firm produces this good using labor supplied by each household. We assume that, for some scalar ϕ that indexes the state of technology, each household has productivity $\omega^h(\phi)$. The firm has a production function

$$Y = \mathbb{E} \left[\omega^h(\phi) \ell_d^h(\phi) \right] \quad (1)$$

where $\ell_d^h(\phi)$ is the labor demanded from household h .

Each household h supplies labor at a wage $w^h(\phi)$ and consumes the final good, whose price we normalize to one. Households have preferences $u^h(c, \ell)$ over consumption c and labor supply ℓ , and they face a non-linear income tax schedule.

Income taxes take the form of a function $T(\cdot, \phi)$ that maps pre-tax income to post-tax income. Taxes are, in general, indexed by the state of technology.

2.2 Equilibrium conditions

For each state of technology ϕ —and its associated productivities $\{\omega^h(\phi)\}$ and tax schedule $T(\cdot, \phi)$ —an equilibrium is a profile of consumption $\{c^h(\phi)\}$, labor supply $\{\ell^h(\phi)\}$, labor supply $\{\ell_d^h(\phi)\}$, production Y , and wages $\{w^h(\phi)\}$ that satisfy the following conditions.

Household optimization requires

$$(c^h(\phi), \ell^h(\phi)) \in \arg \max_{c, \ell} u^h(c, \ell) \quad \text{s.t.} \quad c = w^h(\phi)\ell - T(w^h(\phi)\ell, \phi) \quad (2)$$

Firm optimization requires

$$(Y, \{\ell_d^h(\phi)\}_{h \in \mathcal{H}}) \in \arg \max_{\tilde{Y}, \{\ell_d^h\}_{h \in \mathcal{H}}} \tilde{Y} - \mathbb{E}[w^h(\phi)\ell_d^h] \quad \text{s.t.} \quad \tilde{Y} = \mathbb{E}[\omega^h(\phi)\ell_d^h] \quad (3)$$

Finally, market clearing requires¹

$$Y = \mathbb{E}[c^h(\phi)] \quad \text{and} \quad \ell^h(\phi) = \ell_d^h(\phi) \quad \text{for all } h \in \mathcal{H} \quad (4)$$

Note firm optimization is equivalent to technological feasibility (i.e., Equation 1) plus the statement that for each household's wage equals its productivity, i.e., $w^h(\phi) = \omega^h(\phi)$.

2.3 The Cost of Redistribution

We now define the key concept of the paper: the *cost of redistribution*. In words, the cost of redistribution at any percentile p of the income distribution is the lump-sum tax required to finance a nonlinear tax reform that redistributes income from above p to below p .

To formalize this notion, we introduce some notation. First, we let $z^h(T, \phi)$ denote the income each household h would earn at wages $w^h(\phi) = \omega^h(\phi)$ and taxes T .

$$z^h(T, \phi) = \omega^h(\phi)\ell^h(T, \phi), \quad \text{where} \quad \ell^h(T, \phi) \equiv \arg \max_{\ell} u^h(\omega^h(\phi)\ell - T(\omega^h(\phi)\ell), \ell) \quad (5)$$

Second, we let $\text{Rev}(T, \phi)$ denote the tax revenue raised under taxes T at technology ϕ .

$$\text{Rev}(T, \phi) \equiv \mathbb{E}[T(z^h(T, \phi))] \quad (6)$$

Third, for any percentile $p \in (0, 1)$, we let $\mathcal{Z}(p, \phi)$ denote income at the p^{th} percentile given technology ϕ and taxes $T(\cdot, \phi)$.²

We also introduce two particular tax reforms. First, we let $\bar{\Delta}(\cdot)$ denote the lump-sum

¹Government budget balance is redundant by Walras' law.

²Our formal results use assumptions that guarantee $z^h(T, \phi)$, $\text{Rev}(T, \phi)$, and $\mathcal{Z}(p, \phi)$ are well defined.

tax reform that raises taxes by \$1 at all incomes. Second, we let $\Delta(\cdot, p, \delta p, \phi)$ denote the “redistributive reform” that lowers taxes at all incomes below $\mathcal{Z}(p, \phi)$ by \$1, raises taxes above $\mathcal{Z}(p + \delta p, \phi)$ by $\$p/(1 - p)$, and is linear in between. The left panel of Figure 1 illustrates this reform.

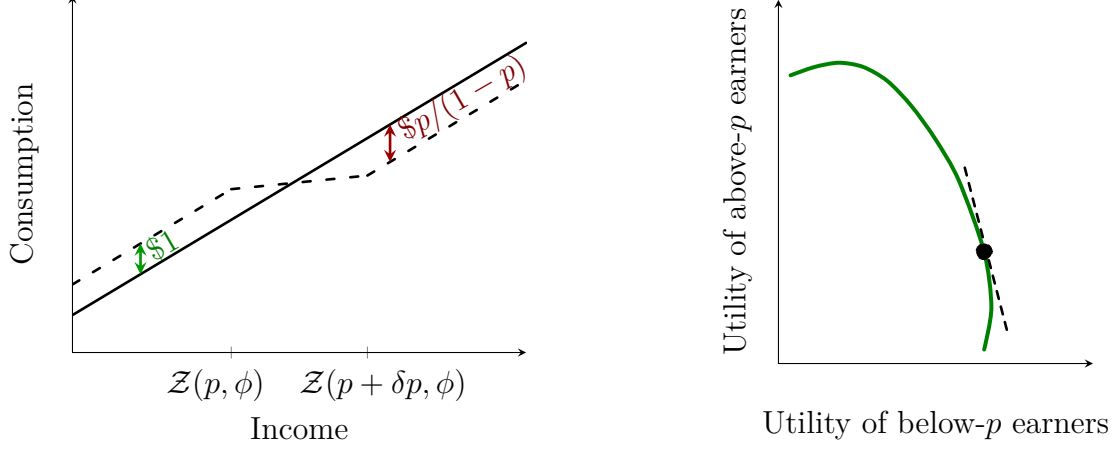


Figure 1: Left panel: Redistributive tax reform. Right panel: Utility possibilities frontier between households above and below income percentile p .

We define the cost of redistribution at percentile p and technology ϕ —denoted $\kappa(p, \phi)$ —as the lump-sum tax required to finance a small redistributive tax reform. Formally,

$$\kappa(p, \phi) \equiv \lim_{\delta p \rightarrow 0^+} \frac{d}{d\epsilon} \bigg|_{\epsilon=0} \min \left\{ k \mid \text{Rev} \left(T(\cdot, \phi) + \epsilon [\Delta(\cdot, p, \delta p, \phi) + k \bar{\Delta}(\cdot)], \phi \right) \geq 0 \right\} \quad (7)$$

In the special case where household labor supply does not respond to lump-sum taxes, $\kappa(p, \phi)$ is simply the fiscal cost of a redistributive tax reform. Since the reform $\Delta(\cdot, p, \delta p, \phi)$ is defined so as to have zero mechanical effects on tax revenue, this cost is equal to a redistributive reform’s behavioral effects on revenue. The general formulation accounts for that, when there are income effects, the lump-sum taxes used to finance a redistributive reform can, themselves, alter behavior and so induce secondary effects on revenue.

The cost of redistribution connects naturally to notions familiar from the income taxation literature. First, it is closely related to the slope of a constrained Pareto frontier. The left panel of Figure 2 depicts—on the x-axis—the average, money-metric utility of households with incomes below percentile p and—on the y-axis—the average money-metric utility of households with incomes above percentile p . The green curve represents the utility possibilities frontier achievable with some income tax schedule. The cost of redistribution provides a measure of this frontier’s slope local to any point on the frontier (i.e. any constrained efficient tax schedule). Concretely, a redistributive reform at percentile p , combined with the increase in lump-sum taxes required to finance it, transfers $1 - \kappa(p, \phi)$ units of consumption to each household below p and transfers $-p/(1 - p) - \kappa(p, \phi)$ to each household above p . The ratio of these two changes,

$$s(p, \phi) = - \frac{p/(1-p) + \kappa(p, \phi)}{1 - \kappa(p, \phi)} \quad (8)$$

is the local slope of the frontier. Intuitively, this slope's magnitude increases monotonically in $\kappa(p, \phi)$; the higher is the cost of redistribution, the more one must lower the utility of above- p households to raise that of below- p households.³

The cost of redistribution also has close parallels in Saez [2001]'s variational approach to optimal income taxation. This approach leverages the necessary condition that, local to an optimum, the welfare costs of a tax reform that raises marginal taxes around an income percentile z should equal its fiscal benefits, i.e., the mechanical revenues generated by higher taxes, holding household behavior fixed, less any additional costs associated with households' behavioral responses. Absent income effects, these trade-offs are captured by

$$\underbrace{\alpha(p, \phi) \varepsilon(p, \phi) \frac{T'(\mathcal{Z}(p, \phi), \phi)}{1 - T'(\mathcal{Z}(p, \phi), \phi)}}_{\text{Behavioral revenue cost}} = \underbrace{\mathbb{E}[1 - \lambda^h(\phi) | z^h > \mathcal{Z}(p, \phi)]}_{\text{Mechanical revenue benefit less welfare cost}} \quad (9)$$

where $\alpha(p, \phi)$ is the first-order Pareto shape of the income distribution at percentile p and $\varepsilon(p, \phi)$ is the average elasticity of taxable income at percentile p .⁴ The right-hand side captures the value of the mechanical income transfer associated with the reform: it takes one dollar from each taxpayer above percentile p —whose consumption the planner values at some $\lambda^h(\phi)$ —and gives it to the government. The left-hand side captures the fiscal externalities associated with the reduced labor supply of households whose marginal taxes rise. Without income effects, these fiscal externalities are equivalent to the cost of redistribution $\kappa(p, \phi)$. Both simply measure the behavioral revenue costs of an increase in marginal taxes.

2.4 Changes in the Cost of Redistribution

This paper's central question is how the cost of redistribution responds to changes in technology. In other words we ask—at each income percentile p —how $\kappa(p, \phi)$ changes with respect to ϕ .

It is important to emphasize that any comparison of the cost of redistribution across technologies ϕ depends on the taxes $T(\cdot, \phi)$ applied at those technologies. The tax schedule is a key determinant of the cost of redistribution because it determines the size of fiscal externalities associated redistributive reforms. In order to highlight the role of techno-

³If the cost of redistribution is above 1 or below $-p/(1-p)$, then the frontier is upward-sloping, i.e. taxes are Pareto inefficient. The former case corresponds to the situation where a redistributive reform is so distortionary that the lump-sum taxes required to finance it are larger than the initial transfers. The latter corresponds to the situation where a redistributive reform actually increases revenue, and in fact raises so much revenue that redistributing this revenue makes even households above p better off.

⁴These quantities are familiar in the income taxation literature. Section 3 defines them formally.

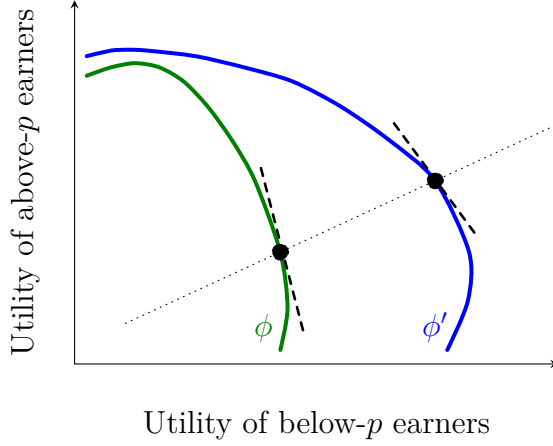


Figure 2: Changes in the cost of redistribution with respect to technology.

logical change itself rather than technology-associated changes in taxes, we assume that taxes are held constant in the sense that the marginal taxes at each percentile of the income distribution do not depend on technology. This assumption provides a practical framework for benchmarking and adjusting the tax schedule, with tax brackets shifting in proportion to income growth at each percentile.⁵

Assumption 1. *At all technologies ϕ and ϕ' , at all percentiles $p \in (0, 1)$*

$$T'(\mathcal{Z}(p, \phi), \phi) = T'(\mathcal{Z}(p, \phi'), \phi'). \quad (10)$$

Figure 2 illustrates the role of Assumption 1 graphically. Starting from some initial tax schedule $T(\cdot, \phi)$ at technology ϕ , technology changes from ϕ to ϕ' , expanding the utility possibilities frontier. How the cost of redistribution changes—i.e. how the frontier’s slope changes—depends on what point on the new frontier one considers. The role of Assumption 1 is to select a particular point on this frontier, namely the one that intersects the dotted line through our starting point under ϕ_0 .

Our goal in emphasizing changes in the cost of redistribution is to provide objective, positive insights free of normative judgments. These insights are nonetheless informative for policymakers, who can combine them with their own views about changes in the normative benefits of redistribution. While we do not emphasize them, our results have straightforward normative implications: For example, if the normative benefit of redistribution—as captured by the weights $\lambda^h(\phi)$ in Equation 9—remains constant while the cost of redistribution falls (rises), then taxes can be improved through progressive

⁵While one could conceivably hold taxes fixed in some alternative way, Assumption 1 has three distinct advantages. First, it is simple and intuitive. Second, compared to the other simple option—holding fixed marginal taxes at each income level, rather than percentile—it has the advantage that the cost of redistribution does not respond to changes in the size of a unit of consumption, as may occur due to inflation. Third, because we consider changes in the cost of redistribution at fixed percentiles of the income distribution rather than at fixed incomes, it is more tractable to assume that marginal taxes are held fixed by percentile rather than income.

(regressive) redistributive reforms.

3 Theoretical Results

Having introduced our paper’s core question—how the cost of redistribution responds to technological changes—we now turn to our theoretical results. We start from a set of three benchmark conditions under which we show the cost of redistribution is technology-invariant. We then relax each condition to illuminate distinct mechanisms through which the cost of redistribution can change. While we focus on the main text on qualitative results that simply *sign* changes in the cost of redistribution, our proofs in Appendix A provide full quantitative expressions.

All of our theoretical results assume that households preferences are quasilinear in consumption and feature a constant elasticity of labor supply. Future drafts will relax these assumptions.

Assumption 2. *For all $h \in \mathcal{H}$, there exist $\theta^h, \beta^h > 0$ such that*

$$u^h(c, \ell) = c - \frac{(\ell/\theta^h)^{1+1/\beta^h}}{1 + 1/\beta^h} \quad (11)$$

Under Assumptions 1 and 2, all changes in cost of redistribution can be expressed in terms of changes in two familiar statistics. First, letting $F(z, \phi)$ denote the CDF of the income distribution at income z and technology ϕ , we denote by

$$\alpha(p, \phi) \equiv -\frac{\partial \log}{\partial \log z} \Big|_{z=\mathcal{Z}(p, \phi)} [1 - F(z, \phi)] \quad (12)$$

the first-order Pareto shape of the income distribution at percentile p . The cost of redistribution is proportional to $\alpha(p, \phi)$, which captures the amount of income that is distorted by a redistributive reform’s change in marginal taxes. Second, letting $\hat{\Delta}(\cdot)$ be the linear function defined by $\hat{\Delta}(z) = z$ and letting $z^h(\phi) \equiv z^h(T(\cdot, \phi), \phi)$, we denote by

$$\varepsilon^h(\phi) \equiv \left(1 - T'(z^h(\phi), \phi)\right) \frac{d}{d\epsilon} \Big|_{\epsilon=0} \log z^h \left(T(\cdot, \phi) - \epsilon \left(\hat{\Delta}(\cdot) - z^h(\phi) \bar{\Delta}(\cdot)\right), \phi\right) \quad (13)$$

household h ’s compensated elasticity of taxable income at technology ϕ .⁶ We let $\varepsilon(p, \phi) \equiv \mathbb{E}[\varepsilon^h(\phi) | z^h(\phi) = \mathcal{Z}(p, \phi)]$ denote the average compensated elasticity of taxable income at income percentile p . The cost of redistribution is proportional to $\varepsilon(p, \phi)$, which captures the responsiveness of households whose marginal taxes rise under a redistributive reform.

⁶A household h ’s compensated elasticity of labor supply, $\varepsilon^h(\phi)$, equals β^h when taxes are linear local to its income. When taxes are locally convex, $\varepsilon^h(\phi)$ exceeds β^h since a reform that lowers marginal taxes at each income not only lowers marginal taxes at h ’s initial income, but also moves her to a new income level where marginal taxes are lower [Scheuer and Werning, 2018].

The cost of redistribution also depends on marginal taxes, but, under Assumption 1, these are held constant at each income percentile p . As a result, the proportional change in the cost of redistribution at any percentile is simply the sum of proportional changes in the first-order local Pareto shape and the compensated elasticity at that percentile.

Lemma 1. *For any income percentile $p \in (0, 1)$ and technologies ϕ and ϕ' ,*

$$\log \kappa(p, \phi') - \log \kappa(p, \phi) = \left[\log \alpha(p, \phi') - \log \alpha(p, \phi) \right] + \left[\log \varepsilon(p, \phi') - \log \varepsilon(p, \phi) \right] \quad (14)$$

3.1 A Neutral Benchmark

We begin our analysis by providing a set of three conditions under which the cost of redistribution is technology-invariant. The following sections relax each of these conditions in turn to characterize three distinct mechanisms through which changes in technology can affect the cost of redistribution away from the benchmark.⁷

Our first condition is that all households experience the same proportional increase in productivity.

Condition 1. *For all households $h, h' \in \mathcal{H}$ and technologies ϕ, ϕ' , we have*

$$\log \omega^h(\phi') - \log \omega^h(\phi) = \log \omega^{h'}(\phi') - \log \omega^{h'}(\phi) \quad (15)$$

Our second condition is that—while household labor supply elasticities may vary within incomes—the distribution of elasticities is the same across all incomes at some initial technology ϕ_0 . Below and throughout, we use $z^h(\phi)$ as shorthand for $z^h(T(\cdot, \phi), \phi)$, i.e. household h 's income under technology ϕ .

Condition 2. *For all incomes z, z' and scalars β ,*

$$\mathbb{P}[\beta^h \leq \beta | z^h(\phi_0) = z] = \mathbb{P}[\beta^h \leq \beta | z^h(\phi_0) = z'] \quad (16)$$

Conversely, our third condition is that—while household labor supply elasticities may vary across incomes—labor supply elasticities do not vary within incomes at ϕ_0 .

Condition 3. *For all incomes z , $\mathbb{V}ar[\beta^h | z^h(\phi_0) = z] = 0$.*

Together, these three conditions guarantee that the cost of redistribution is technology-invariant at every income percentile.

Proposition 1. *If Conditions 1–3 hold, then for all p and all ϕ_1 , $\kappa(p, \phi_0) = \kappa(p, \phi_1)$.*

⁷In order to highlight qualitative mechanisms, the main text focuses on cases where only one of these benchmark conditions is violated at a time.

By Lemma 1, one way for Proposition 1 to be true is if both the first-order local Pareto shape $\alpha(p, \phi)$ and the local mean compensated elasticity $\varepsilon(p, \phi)$ are the same at technologies ϕ_0 and ϕ_1 . We now explain why this is the case.

First, consider why the first-order local Pareto shape is technology-invariant. Under Conditions 2 and 3, all households share a common elasticity of labor supply disutility, $1 + 1/\beta$. As a result, their labor supply decisions are described by similar first-order conditions:

$$\log z^h(\phi) = (1 + \beta) \log \omega^h(\phi) + \beta \log (1 - T'(z^h(\phi), \phi)) \quad (17)$$

Condition 1 states that all households h experience the same change in $\log \omega^h(\phi)$, and Assumption 1 ensures that taxes are indexed to technology such that all households h experience no change in $T'(z^h(\phi), \phi)$.⁸ As a result, Equation 17 implies that all households experience the same proportional increase in income. In other words, technology shifts the distribution of log income uniformly up or down. This has no impact on the distribution's first-order Pareto shape at percentile p , which measures the sensitivity of the number of earners above a log-income level to that log-income level, local to the log-income level $\log Z(p, \phi)$.

Second, consider why the local mean compensated elasticity is technology-invariant. Our notion of each household h 's compensated elasticity depends both on the preference parameter β^h and the local curvature of the tax schedule, $\partial \log(1 - T'(z, \phi)) / \partial \log z|_{z=z^h(\phi)}$ (see discussion in Footnote 6). Since under Conditions 2 and 3 all households have the same $\beta^h = \beta$, elasticities can change only through the tax curvature term. This term can conveniently be expressed as

$$\frac{\partial \log(1 - T'(z, \phi))}{\partial \log z} \Big|_{z=Z(p, \phi)} = \frac{d \log (1 - T'(Z(p, \phi), \phi))}{dp} \left[\frac{\partial \log Z(p, \phi)}{\partial p} \right]^{-1} \quad (18)$$

Assumption 1 ensures taxes are indexed in such a way that the first term on the right-hand side is technology-invariant. Since, as discussed above, Conditions 1–3 imply that all households' log incomes change the same amount in response to technology, the second term on the left-hand side is technology-invariant as well. As a result, the average elasticity at every percentile stays the same.

Since technology affects neither the first-order Pareto shape nor the average elasticity at any income percentile p , the cost of redistribution at every income percentile is technology-invariant.

⁸In order for Assumption 1 to ensure that all households h experience no change in $T'(z^h(\phi), \phi)$, it must be that households' income ranks are also technology invariant. In the proof, we provide technical conditions that guarantee this.

3.2 Non-uniform changes in wages

In practice, technological changes may have different effects on the wages of households in different parts of the income distribution. For example, the automation of cognitive routine jobs may “hollow out” the middle of income distribution. Other technological changes could increase inequality throughout the income distribution.

We now study the effect that technology’s heterogeneous wage impacts have on the cost of redistribution. To do so, we maintain Conditions 2 and 3, so that elasticities are uniform both within and across income levels, but relax Condition 1. We show in this case that the cost of redistribution at each percentile of the income distribution changes depending on whether wage inequality is locally increasing or decreasing.

Proposition 2. *If Conditions (2) and (3) hold, then for all $p \in (0, 1)$,*

$$\frac{\partial \kappa(p, \phi_0)}{\partial \phi_0} < 0 \quad \Longleftrightarrow \quad \frac{d}{dp} \mathbb{E} \left[\frac{d \log \omega^h(\phi_0)}{d \phi_0} \mid z^h(\phi_0) = \mathcal{Z}(p, \phi_0) \right] > 0. \quad (19)$$

A simple way to understand this result is to notice that if proportional wage gains from changing technology increase as a function of p , higher-income households experience larger income gains. This stretching of the income distribution manifests in a decrease in $\alpha(p, \phi)$ and hence a decrease in the cost of redistribution. Economically, this corresponds to the observation that the further income percentile $p + \delta p$ is from percentile p , the less marginal taxes are raised between them by the redistributive reform $\Delta(\cdot, p, \delta p, \phi)$. When the distance between percentiles p and $p + \delta p$ grows in logs, taxes flatten by more than incomes rise, so the total distortion of a redistributive reform is smaller.

If marginal taxes are constant local to percentile p , this stretching effect is the only impact of a change in technology. If not, the widening of the income distribution while marginal taxes are held fixed at each income percentile changes the curvature of marginal taxes as function of income, implying that elasticities $\varepsilon(p, \phi)$ at each percentile p change as well. However, we prove that the stretching effect always dominates this secondary tax-curvature effect.

Example 1. *Suppose that technological change increases wage inequality throughout the income distribution. I.e., higher-income households experience larger proportional wage increases. Proposition 2 then implies that the cost of redistribution falls at all income percentiles. If the relative social value of consumption at each income percentile is unchanged, then the reduced cost of redistribution implies there are first-order welfare gains from increasing marginal taxes at each income percentile.*

Example 2. *Suppose that the automation of cognitive routine jobs “hollows out” the income distribution, reducing the wages of middle-income earners. I.e., below the middle of the income distribution, higher-income households experience larger wage losses and above*

the middle of the income distribution, higher-income households experience smaller wage losses. Proposition 2 then implies that the cost of redistribution rises at income percentiles below the median and falls at income percentiles above the median. Absent changes in the relative social value of consumption at each income percentile, these changes in the cost of redistribution imply there are first-order gains from lower taxes on below-median earners and higher taxes on above-median earners.

3.3 Elasticity variation across incomes

The neutral benchmark in Proposition 1 assumes there is no variation in elasticities across income levels. In fact, elasticities of taxable income out of tax changes (ETIs) do vary with income. For example, [Becko and Sztutman \[2024\]](#) estimate that ETIs in the US are highest at low incomes (around 0.5) lowest at moderate incomes (around 0.2), and higher again at high incomes (around 0.35) (see also [Gruber and Saez \[2002\]](#) and [Kumar and Liang \[2020\]](#)).

We now study the effect of elasticity variation across incomes on changes in the cost of redistribution. To do so, we maintain Conditions 1 and 3—so that productivity changes are uniform and elasticities are uniform within income levels—but relax Condition 2. We show in this case that the cost of redistribution at each percentile of the income distribution changes depending on whether elasticities are locally increasing or decreasing in income.

Proposition 3. *If Conditions 1 and 3 hold and productivities are increasing in ϕ , then for all $p \in (0, 1)$,*

$$\frac{\partial \kappa(p, \phi_0)}{\partial \phi_0} < 0 \quad \Longleftrightarrow \quad \frac{d}{dp} \mathbb{E} [\beta^h \mid z^h(\phi_0) = \mathcal{Z}(p, \phi_0)] > 0 \quad (20)$$

The economic mechanism behind Proposition 3 is similar to that of Proposition 2. If higher income households are more elastic, then they respond by more than lower income households to a uniform increase in wages. This stretches out the income distribution, and in particular reduces the marginal tax hike associated with a redistributive reform by more than it raises incomes. As a result, the negative fiscal externalities associated with such reforms decrease.⁹ As in the case of Proposition 2, such stretching of the income distribution not only lowers the first-order local Pareto parameter $\alpha(p, \phi)$ but can also

⁹It makes no difference for the cost of redistribution why the income distribution stretches out (so long as $\varepsilon(p, \phi)$ is the same). Mathematically, differentiating households' first-order condition implies that

$$\frac{d \log z^h(\phi_0)}{d \phi_0} = (1 + \beta^h) \frac{d \log w^h(\phi_0)}{d \phi_0} \quad (21)$$

So higher income households experience weakly larger income growth so long as both β^h s and $d \log w^h(\phi_0)/d \phi_0$ are weakly larger for households h with higher incomes.

change the local average elasticity $\varepsilon(p, \phi)$ when taxes are nonlinear local to percentile p . However, we show that the first force always dominates.

Example 3. *Suppose that—as discussed above—income-conditional average ETIs are U-shaped as a function of income percentiles, reaching a minimum at some income percentile $\underline{p}$. Then if productivity increases uniformly, the cost of redistribution rises below percentile $\underline{p}$ and falls above percentile $\underline{p}$. Absent changes in the relative social value of consumption at each income percentile, these changes in the cost of redistribution imply there are first-order gains from decreasing marginal taxes below $\underline{p}$ and raising them above $\underline{p}$.*

3.4 Elasticity variation within incomes

The neutral benchmark in Proposition 1 assumes all households with the same income have the same elasticity. In fact, elasticities do vary conditional on income. [Becko and Sztutman \[2024\]](#) estimate income-conditional ETI variances by several different methods, providing a lower bound of 0.2 and point estimates as high as 1.5.

We now study the effect of elasticity variation within incomes on changes in the cost of redistribution. To do so, we maintain Conditions 1 and 2, so that productivity changes are uniform and elasticities have the same distribution across income levels, but relax Condition 3. We show in this case that the impact of productivity growth on the cost of redistribution depends critically on the local shape of the income distribution.

We refer to the shape parameter that mediates the impacts of productivity improvements under within-income elasticity heterogeneity as the second-order local Pareto shape of the income distribution [[Jacobs et al., 2017](#), [Hendren, 2020](#)]. It captures how quickly the aggregate income earned at any income level falls with that income level. Formally, we denote

$$\tilde{\alpha}(p, \phi) = - \frac{d \log(zf(z, \phi))}{d \log z} \Big|_{z=\mathcal{Z}(p, \phi)} \quad (22)$$

where recall $f(\cdot, \phi)$ is the income density at technology ϕ . When the income distribution is Pareto, the first- and second-order local Pareto shapes are constant across income percentiles and both equal the distribution’s Pareto parameter.¹⁰

For simplicity, we state the result in the case where income taxes are affine.

Proposition 4. *Suppose Conditions 1 and 2 hold, Condition 3 fails, the tax schedule $T(\cdot, \phi_0)$ is affine, and productivities increase with ϕ . Then for all $p \in (0, 1)$,*

$$\frac{\partial \kappa(p, \phi_0)}{\partial \phi_0} > 0 \quad \Longleftrightarrow \quad \tilde{\alpha}(p, \phi_0) > 0 \quad (24)$$

¹⁰More generally, one can show that

$$\tilde{\alpha}(p, \phi) = \alpha(p, \phi) - \frac{\partial \log \alpha(p, \phi)}{\partial p} \left[\frac{\partial \log \mathcal{Z}(p, \phi)}{\partial p} \right]^{-1} \quad (23)$$

In the two previous propositions, we showed that cross-income variation in either productivity gains or disutilities of labor supply can change the cost of redistribution by stretching or compressing the income distribution, raising or lowering the Pareto parameter $\alpha(p, \phi)$. Proposition 4 turns off both of these channels. As we formally verify in the proof, this implies that technological changes have no first-order impact on $\alpha(p, \phi)$. By Lemma 1, the cost of redistribution therefore changes one-for-one with average income-conditional elasticities $\varepsilon(p, \phi)$. Since we have assumed taxes are linear, each household's compensated elasticity $\varepsilon^h(\phi)$ is simply its preference elasticity β^h , so any changes in these income-conditional elasticities—and hence the cost of redistribution—must reflect changes in the β^h -composition of income percentiles.

Why then, in the presence of within-income elasticity heterogeneity, do productivity improvements change the elasticity composition of the income distribution? This is best explained with the help of a picture. Figure 3 shows the elasticity composition of the income distribution in a simple example, before and after a uniform productivity improvement. We assume the population contains an equal proportion of low- and high-elasticity types ($\beta^h = 0$ and $\beta^h = 1$) at every income level, as shown in the top panel of the figure. The average elasticity $\varepsilon(p, \phi_0)$ is therefore equal to $1/2$ at every income percentile p .

Next, consider how the income distribution changes following a uniform productivity improvement. The distribution shifts up, as shown in the bottom panel of Figure 3, since all households earn more when their wages rise. However, this effect is larger for households with $\beta^h = 1$, since they earn more not only due to higher wages at their initial labor supply but also because they respond to these higher wages by working more hours. The log-income distribution of high-elasticity households therefore shifts up by more than that of low-elasticity households.¹¹ As a result, in parts of the income distribution where the log-income density is rising, higher income households come to comprise a larger share of earners, so that the average elasticity rises. The reverse is true at incomes where the log-income density is falling. Note that the log income density $\partial F(z, \phi) / \partial \log z$ is equal to $zf(z, \phi)$, so the second-order Pareto shape at p is positive if and only if the log-income density is falling. Such parts of the income distribution (such as the top) become disproportionately populated with higher-income households following uniform productivity increases, raising the cost of redistribution at these percentiles.

Example 4. *Suppose—in line with the estimates of Hendren [2020] and Becko and Sztutman [2024]—that the local Pareto shape of the income distribution rises with income, crossing from negative to positive around the 60th percentile. Then if productivity increases uniformly, the cost of redistribution falls below the 60th percentile and rises above it. Absent changes in the relative social value of consumption at each income percentile, these changes in the cost of redistribution imply there are first-order gains from lowering*

¹¹Working with the log, rather than level, or income implies that the income distribution of each group simply shifts up, rather than also spreading out.

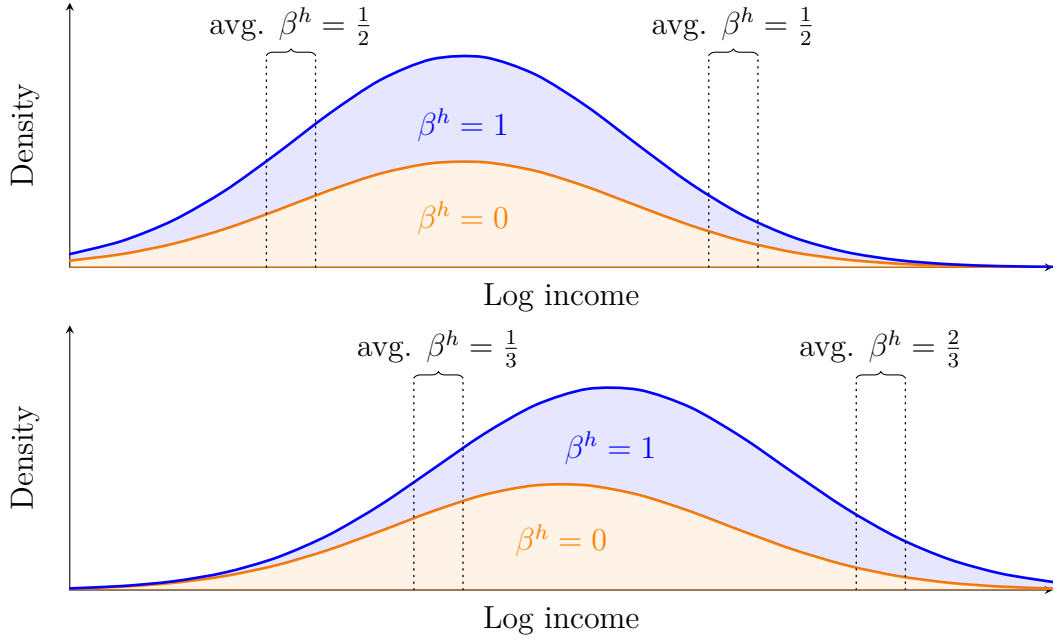


Figure 3: Distribution of log income by elasticity group, before and after a uniform productivity increase.

marginal taxes below the 60th percentile and raising taxes above it.

4 Tax Avoidance

In our baseline model from Section 2, each household's labor supply is determined by the relationship between its labor supply and real income. This implies its elasticity of taxable income out of wage changes is simply one plus its elasticity of taxable income out of tax changes.¹² This is essential for the sorting mechanism emphasized by Proposition 4, as it implies that the same households who sort to the top of the income distribution in response to productivity improvements are also the ones who respond the most to tax reforms.

But are the most wage-elastic households the most tax-elastic in practice? This is ultimately an empirical question, and one that we answer quantitatively in Section 5. This section expands our theoretical framework and so that it can accommodate differences between wage elasticities and tax elasticities in a model-consistent way, and it extends our results accordingly.

To microfound differences between wage elasticities and tax elasticities, we allow households to report taxable income different than their earned income. We do not take a stance on whether households *avoid* taxes within legal limits—say, by itemizing deductions or arranging to receive their wages as capital income—or illegally *evade* them.

¹²Wage elasticities mechanically exceed tax elasticities by one, since wage raises increase post-tax income even if households do not adjust their labor supply.

Below, we explain how avoidance and/or evasion rationalize elasticity differences, and we show how our results generalize in their presence.

4.1 Model of tax avoidance

We augment the baseline model by assuming that households can separately choose the income $I = w^h(\phi)\ell$ that they earn and the income z that they report for tax purposes. However, they experience a disutility cost $v^h(z, I)$ from reporting different income than they earn, which can represent time spent finding tax deductions (in the case of avoidance) or fear of a tax audit (in the case of evasion).¹³ We therefore replace the household optimization condition in Equation 2 with

$$(c^h(\phi), \ell^h(\phi), z^h(\phi)) \in \arg \max_{c, \ell, z} u^h(c, \ell) - v^h(z, w^h(\phi)\ell) \quad \text{s.t.} \quad c = w^h(\phi)\ell - T(z, \phi) \quad (25)$$

Away from the baseline case where $v^h(z, I) = -\infty \mathbb{1}_{z=I}$, this model predicts that households can respond differently changes in wages and changes in marginal taxes. Why? A wage raises and a tax reduction have identical effects on a household's incentives to work *holding fixed* the share of its income that it reports as taxable. However, a tax reduction also weakens the household's incentives to avoid or evade taxes, so it may raise reported income by more than a wage raise (ignoring the wage raise's mechanical effect on income). Importantly, if households differ in their ability or willingness to avoid or evade taxes, then the most wage-elastic households need not be the most tax-elastic.

4.2 The cost of redistribution with tax avoidance

We now revisit the results of Section 3 in the presence of tax avoidance. We define the cost of redistribution at percentile p and technology ϕ , $\kappa(p; \phi)$, in the same way as before: the fiscal cost of a tax reform that raises retention by one dollar below p , financed by a uniform increase in taxes above p . This cost depends only households' elasticities of *reported income*. Note that, despite households' more complex choice of income with misreporting, the redistributive benefit this reform is the same as before: Each household's first-order change in utility from a reform still equals the change in retention at their initial reported income, by the envelope theorem [Feldstein, 1999].

In order to compare our results with and without tax avoidance, we introduce a few pieces of notation. First, we continue to let $\varepsilon_{1-T'}^h(\phi)$ denote household h 's taxable (reported) income supply elasticity with respect to changes in $(1 - \text{marginal tax rate})$. Second, we let $\varepsilon_w^h(\phi)$ denote h 's taxable income elasticity with respect to changes in wages.

¹³This simple formulation ignores any revenues associated with audits that identify misreporting and recoup tax payments [Chetty, 2009]. As such, it can be thought as providing an extreme case where the two elasticity concepts are the most distinct.

Third and finally, we let $\kappa^{\text{base}}(p, \phi)$ denote what would be the cost of redistribution for a hypothetical population with the true distribution of reported incomes $\{z^h(\phi)\}$ and the true elasticities $\{\varepsilon_{1-T'}^h(\phi)\}$ out of tax changes, but counterfactual elasticities $\{\tilde{\varepsilon}_w^h(\phi)\}$ out of wage changes. Specifically, we consider the hypothetical population with wage elasticities $\tilde{\varepsilon}_w^h(\phi)$ equal to the true level of tax elasticities, i.e. $\tilde{\varepsilon}_w^h(\phi) = \varepsilon_{1-T'}^h(\phi)$.

Our next result illustrates the role of misreporting by comparing these actual and hypothetical costs of redistribution. For simplicity, we focus on the case where taxes are affine.

Proposition 5. *Suppose taxes $T(\cdot, \phi_0)$ are affine. For all $p \in (0, 1)$, the following are true:*

- *If Conditions 2 and 3 hold, then*

$$\frac{d \log \kappa(p, \phi_0)}{d \phi_0} = \frac{\mathbb{E} \left[\varepsilon_w^h(\phi_0) \mid z^h(\phi_0) = \mathcal{Z}(p, \phi_0) \right]}{\mathbb{E} \left[\varepsilon_{1-T'}^h(\phi_0) \mid z^h(\phi_0) = \mathcal{Z}(p, \phi_0) \right]} \frac{d \log \kappa^{\text{base}}(p, \phi_0)}{d \phi_0} \quad (26)$$

- *If Conditions 1 and 3 hold, then*

$$\frac{d \log \kappa(p, \phi_0)}{d \phi_0} = \frac{\frac{d}{dz} \Big|_{z=\mathcal{Z}(p, \phi_0)} \mathbb{E} \left[\varepsilon_w^h(\phi_0) \mid z^h(\phi_0) = z \right]}{\frac{d}{dz} \Big|_{z=\mathcal{Z}(p, \phi_0)} \mathbb{E} \left[\varepsilon_{1-T'}^h(\phi_0) \mid z^h(\phi_0) = z \right]} \frac{d \log \kappa^{\text{base}}(p, \phi_0)}{d \phi_0} \quad (27)$$

- *If Conditions 1 and 2 hold, then*

$$\frac{d \log \kappa(p, \phi_0)}{d \phi_0} = \frac{\text{Cov} \left[\varepsilon_{1-T'}^h(\phi_0), \varepsilon_w^h(\phi_0) \mid z^h(\phi_0) = \mathcal{Z}(p, \phi_0) \right]}{\text{Var} \left[\varepsilon_{1-T'}^h(\phi_0) \mid z^h(\phi_0) = \mathcal{Z}(p, \phi_0) \right]} \frac{d \log \kappa^{\text{base}}(p, \phi_0)}{d \phi_0} \quad (28)$$

This result clarifies exactly how tax avoidance affects our theoretical results. Propositions 2 and 3 change in a relatively limited way: They must simply be adjusted to account for the fact that wage changes stretch out the income distribution according to local wage elasticities, rather than local tax elasticities. The adjustment to Proposition 4 is more qualitative. Intuitively, it accounts for a key assumption underlying the result: that the same households that wage who sort to the top of the income distribution following wage increases are also the most tax elastic. Since sorting concerns wage elasticities and responses to redistribution concern tax elasticities, the assumption need no longer hold when wage and tax elasticities are not equal. Proposition 5 shows that one can exactly account for the difference between wage and tax elasticities simply by considering their covariance rather than their variance. This covariance is therefore a key object of empirical interest, one we will measure Section 5.

5 Empirics

As discussed in Section 4, the strength of the *sorting* mechanism hinges on the within-income correlation of households’ elasticities of taxable income out of wage changes and tax changes. We now turn to evaluating this income-conditional correlation using data from the Survey of Income and Program Participation (SIPP). For brevity, we will refer to elasticities of taxable income out of wages and $(1 - \text{marginal tax rates})$ changes as *wage elasticities* and *tax elasticities*, respectively.

In our theory section, we imagine households choosing a single pre-tax income z , which is, in turn, taxed by a single tax schedule. In reality, different sources of income are taxed differently. We abstract from those differences and focus on changes in a standard measure of households’ pre-tax income in response to changes in their hourly wages and the marginal tax rate on their wage income, following Saez [2001] and Kumar and Liang [2020], among others.

5.1 SIPP Data

SIPP is a comprehensive rotating panel survey administered by the Census Bureau with detailed information on individual earnings, industry, hours worked, demographics, and other relevant dimensions. We use core panels from 2000 to 2013, the last year prior to SIPP’s major redesign in 2014, to ensure that our variable definitions are consistent over time. We link couples within SIPP data.

Our definition of taxable income includes labor income, interest and dividend income, pension, unemployment insurance, business income, property income, and non-property income. Given the absence of exemptions and deductions in SIPP data, our definition closely tracks the notion of broad income in Gruber and Saez [2002]’s. We compute simulated changes in marginal tax rates using NBER’s TAXSIM calculator by Feenberg and Coutts [1993]. Appendix C.3 outlines the data construction process and how we map SIPP variables to TAXSIM inputs.

Our sample selection criteria follow the literature [Kumar and Liang, 2020, Gruber and Saez, 2002]. We are interested in intensive margin responses and hence exclude individuals with taxable income below \$5,000 in 2008 dollars. We also restrict our attention to individuals whose marital status does not change between the years over which our analysis studies changes in their incomes; this avoids tax changes driven by household marital decisions rather than government tax reforms. Lastly, we remove a small number of outliers with marginal tax rates outside of $[-1, 1]$ or a three-year absolute change in marginal taxes exceeding 0.4. Our variation in marginal taxes comes from both federal and state tax changes. In Appendix C.1, we show that across states and over the years, a non-trivial fraction of our sample experienced changes in marginal rates exceeding 5%.

5.2 Empirical approach to estimating elasticities

As a “warm-up” for our estimates of income-conditional elasticity correlation, it is helpful to begin by describing our approach to estimating tax and wage elasticities in the aggregate. Our later procedure (see Section 5.3) applies similar techniques within sub-populations selected by a generalized random forest.

5.2.1 Tax Elasticity

We regress three-year changes in individual taxable income on three-year changes in $(1 - \text{marginal tax rate})$. Following Gruber and Saez [2002], we instrument for marginal tax changes using TAXSIM-simulated three-year ahead changes in marginal tax rates at each individual’s base year earnings profile. Our regression equation is

$$\Delta \log z_{i,t+3} = \alpha + \beta \Delta \log(1 - T')_{i,t+3} + \delta_t + \gamma M_{i,t} + \sum_{j=1}^{10} \theta_j \text{SPLINE}_j(z_{i,t}) + \epsilon_{i,t} \quad (29)$$

Here the Δ operator takes the difference with the three-year lagged outcome of the same individual. On the LHS, $z_{i,t}$ denotes taxable income. On the RHS, $1 - T'$ denotes one minus the marginal tax rate, δ_t is a year fixed effect, and $M_{i,t}$ is a dummy for being married. We also include linear splines of the base year income.

Estimating the tax elasticity, β , requires an instrument for tax changes, since any positive income shocks reduces marginal retention mechanically due to the progressivity of the tax schedule. As in Gruber and Saez [2002], we instrument for $\Delta \log(1 - T')_{i,t+3}$ using $\Delta \log(\widehat{1 - T'})_{i,t+3} \equiv (\widehat{1 - T'})_{i,t+3} - (1 - T')_{i,t}$, where $\widehat{1 - T'}$ is the marginal retention computed by TAXSIM according to tax laws at $t + 3$ but using i ’s earnings profile at t .

5.2.2 Wage Elasticity

We take an analogous approach to estimate wage elasticities. We begin with the regression equation

$$\Delta \log z_{i,t+3} = \alpha + \varepsilon_w \Delta \log w_{i,t+3} + \delta_t + \gamma M_{i,t} + \sum_{j=1}^{10} \theta_j \text{SPLINE}_j(z_{i,t}) + \epsilon_{i,t} \quad (30)$$

where $w_{i,t}$ denotes hourly wage and all other variables are as in Equation 29.¹⁴

In order to estimate ε_r , we must grapple with identification concerns related to the endogeneity of $\Delta \log w_{i,t+3}$. If individuals experience idiosyncratic labor supply shocks, this may cause correlated movements in income and wages. This can occur either if workers have specialized skills such that they are large relative to their effective labor

¹⁴Recall that, without tax avoidance or evasion, $\varepsilon_w = 1 + \varepsilon_r$. The additional factor of one accounts for that wage increases mechanically increase taxable income, even if households do not adjust the hours they work.

markets, or if they change the type of labor they supply (for example, opting for more flexible remote-work arrangements). We address individual-level labor supply shocks by instrumenting for changes in individual wages with the leave-one-out average wage change of other individuals who work in the same industry in the base year. We construct the instrument using the average wage changes of individuals, rather than industries, since the latter could evolve over time due to compositional changes unrelated to productivity innovations. A second concern is that there may be market-level, rather than idiosyncratic, shocks to labor supply. We abstract from this issue by assuming that market-level labor supply shocks are much smaller than market-level labor demand shocks, at least after controlling for time fixed effects.

Concretely, our instrument for wage changes is the following. Let $I_{i,t}$ and $H_{i,t}$ denote i 's income earned and hours worked at her primary job in year t . Let $\mathcal{C}(i, t)$ denote the cohort of individuals who work in the same industry as i at t . The leave-one-out instrument is given by

$$\Delta \hat{w}_{i,t+3} \equiv \underbrace{\log\left(\frac{\sum_{j \in \mathcal{C}(i,t) \setminus i} I_{j,t+3}}{\sum_{j \in \mathcal{C}(i,t) \setminus i} H_{j,t+3}}\right)}_{\text{leave-one-out hourly wage of the cohort at } t+3} - \underbrace{\log\left(\frac{\sum_{j \in \mathcal{C}(i,t) \setminus i} I_{j,t}}{\sum_{j \in \mathcal{C}(i,t) \setminus i} H_{j,t}}\right)}_{\text{leave-one-out hourly wage of the cohort at } t} \quad (31)$$

5.3 Estimating income-conditional correlation

We now turn to estimating the income-conditional correlation of tax and wage elasticities. The premise of our approach is that, provided most of the variation in elasticities can be captured by observable characteristics, correlations in elasticities in the population are similar to correlations in mean elasticities across observable groups. We operationalize this idea in two steps. First, we use machine learning techniques to select the observable groups that best predict households' responses to tax and wage changes. Second, we study the correlation of tax and wage elasticities across these groups.

5.3.1 Estimating heterogeneous elasticities using instrumental forests

Generalized random forests are a flexible tool for estimating heterogeneous treatment effects that are identified through local moment conditions [Athey et al., 2019]. We apply the “instrumental forests” procedure discussed in Section 7 of Athey et al. [2019] to estimate wage and tax elasticities as functions of observable covariates. At a high level, instrumental forests are a method for estimating structural models of the form

$$Y_i = \mu(X_i) + \tau(X_i)W_i + \varepsilon_i \quad (32)$$

Here, Y_i is the outcome of interest, X_i is a vector of covariates, and W_i is the treatment of interest, which may be endogenous. Instrumental forests are a method to estimate

$\tau(\cdot)$ when the researcher has access to a valid instrument Z_i . The instrumental forest method does this in two steps. First, the algorithm greedily grows tree-based predictors by iteratively splitting the subsample along whichever covariate most reduces a measure of loss based on the moment conditions associated with the instrument’s validity. Next, the algorithm estimates $\tau(\cdot)$ by minimizing a weighted version of this moment condition, with weights determined by how frequently observations fall into each terminal node in the previous step. See Section 2 of [Athey et al. \[2019\]](#) for additional details of the algorithm.

We apply the instrumental forest method twice, once to estimate tax elasticities $\beta(X_{i,t})$ as a function of covariates $X_{i,t}$ and once to estimate wage elasticities $\varepsilon_w(X_{i,t})$ as a function of covariates $X_{i,t}$. We take $X_{i,t}$ to be a large set of covariates capturing demographic as well as tax- and job-related characteristics. The demographic variables are sex, marital status, number of dependents, education, and age. The job-related variables are years at current job and occupation category. Tax-related variables include taxable income, marginal rate, self-employment income, mortgage interest, rent, and property tax. We measure all covariates in the base year. These variables capture various dimensions of elasticity heterogeneity—many of which only imperfectly targeted by the tax code—that could account for the within-income elasticity heterogeneity documented in [Becko and Sztutman \[2024\]](#). When we center our estimates prior to applying instrumental forests, we control for marital status, state-year fixed effects, and linear splines of base year income. Details are in [Appendix C.4](#).

Given our estimates $\hat{\beta}(X_i)$ and $\hat{\varepsilon}_w(X_i)$ of the tax and wage elasticity functions, we assign estimated elasticities to each observation in our sample. That is,

$$\hat{\beta}_{i,t} = \hat{\beta}(X_{i,t}) \quad \text{and} \quad \hat{\varepsilon}_{w,i,t} = \hat{\varepsilon}_w(X_{i,t}) \quad (33)$$

[Figure 4](#) displays the joint distribution of these estimates. The x-axis shows each individual’s estimated wage elasticities, minus one. The y-axis shows its tax elasticity. The shading of each cell of the figure represents the number of individuals for which our estimates of wage and tax elasticities are as indicated on the axes. The figure illustrates a few key facts about our estimates. First, both our estimates of wage and tax elasticities are broadly positive, as expected. To the extent that some estimates are negative, this may simply reflect statistical noise, as each group is composed of a relatively small number of households. Second, our estimates of wage and tax elasticities are strongly correlated, with an unconditional correlation of 0.64. This is a first indication that although wage and tax elasticities may not be perfectly aligned, they are closely related.

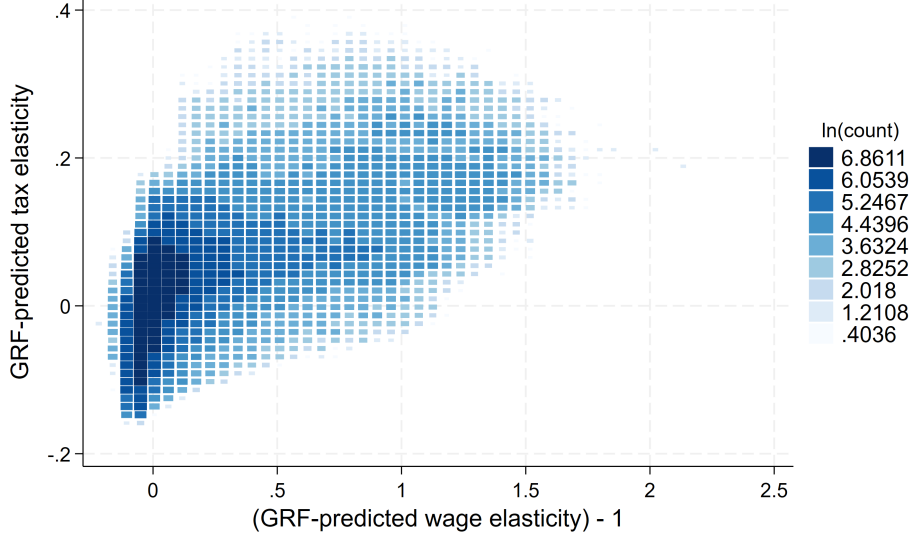


Figure 4: Heatmap of instrumental-forest-predicted wage and tax elasticities.

5.3.2 From heterogeneous estimates to income-conditional correlation

In Section 4.2, we showed that when retention and wage elasticities differ, the relevant statistic to characterize the cost of redistribution is their income-conditional correlation. Formally, we would like to estimate, for each income z , the correlation¹⁵

$$\psi(z) \equiv \text{Corr}[\beta^h, \varepsilon_w^h | z^h = z] = \frac{\text{Cov}[\beta^h, \varepsilon_w^h | z^h = z]}{\text{SD}[\beta^h | z^h = z] \text{SD}[\varepsilon_w^h | z^h = z]} \quad (34)$$

We form an estimate $\hat{\psi}(z)$ of this income-conditional correlation as follows. First, we shift our focus to year-demeaned log incomes $\tilde{z}$ rather than basic incomes z to filter out business cycle variation in the income distribution. Second, we estimate conditional average wage and tax elasticities as functions $\hat{f}_\beta(\cdot)$ and $\hat{f}_{\varepsilon_w}(\cdot)$ of year-demeaned log-income using the local linear regressions

$$\hat{\beta}_{i,t} = f_\beta(\tilde{z}_{i,t}) + \epsilon_{i,t}^\beta \quad \text{and} \quad \hat{\varepsilon}_{w,i,t} = f_{\varepsilon_w}(\tilde{z}_{i,t}) + \epsilon_{i,t}^{\varepsilon_w} \quad (35)$$

where $\hat{\beta}_{i,t}$ and $\hat{\varepsilon}_{w,i,t}$ are the individual-level estimate estimates from Equation 33. Third, we estimate conditional covariances and variances of wage and tax elasticities as functions

¹⁵The correlation in Equation 34 appears slightly different from the one that appears in Proposition 5, since it uses the preference elasticity β^h rather than the taxable income elasticity $\varepsilon_{1-T'}^h$. This difference is immaterial for our quantification in Section 6, since there we focus on top incomes where marginal tax rates are constant and the two elasticity concepts therefore coincide.

$\hat{f}_c(\cdot)$, $\hat{f}_{v_\beta}(\cdot)$, and $\hat{f}_{v_{\varepsilon_w}}(\cdot)$ of year-demeaned log income using the local linear regressions

$$\begin{aligned} (\hat{\beta}_{i,t} - \hat{f}_\beta(\tilde{z}_{i,t})) (\hat{\varepsilon}_{w,i,t} - \hat{f}_{\varepsilon_w}(\tilde{z}_{i,t})) &= \hat{f}_c(\tilde{z}_{i,t}) + \epsilon_{i,t}^c \\ (\hat{\beta}_{i,t} - \hat{f}_\beta(\tilde{z}_{i,t}))^2 &= \hat{f}_{v_\beta}(\tilde{z}_{i,t}) + \epsilon_{i,t}^{v_\beta} \\ (\hat{\varepsilon}_{w,i,t} - \hat{f}_{\varepsilon_w}(\tilde{z}_{i,t}))^2 &= \hat{f}_{v_{\varepsilon_w}}(\tilde{z}_{i,t}) + \epsilon_{i,t}^{v_{\varepsilon_w}} \end{aligned} \quad (36)$$

Fourth, we define $\hat{\psi}(\cdot)$ as the implied covariance function,

$$\hat{\psi}(\tilde{z}) = \hat{f}_c(\tilde{z}) / \left(\sqrt{\hat{f}_{v_\beta}(\tilde{z})} \sqrt{\hat{f}_{v_{\varepsilon_w}}(\tilde{z})} \right) \quad (37)$$

Fifth and finally, we bootstrap our entire procedure (including instrumental forest estimation) to bootstrap-debias our estimates and to form bootstrap confidence intervals. For further details, see Appendix C.5.

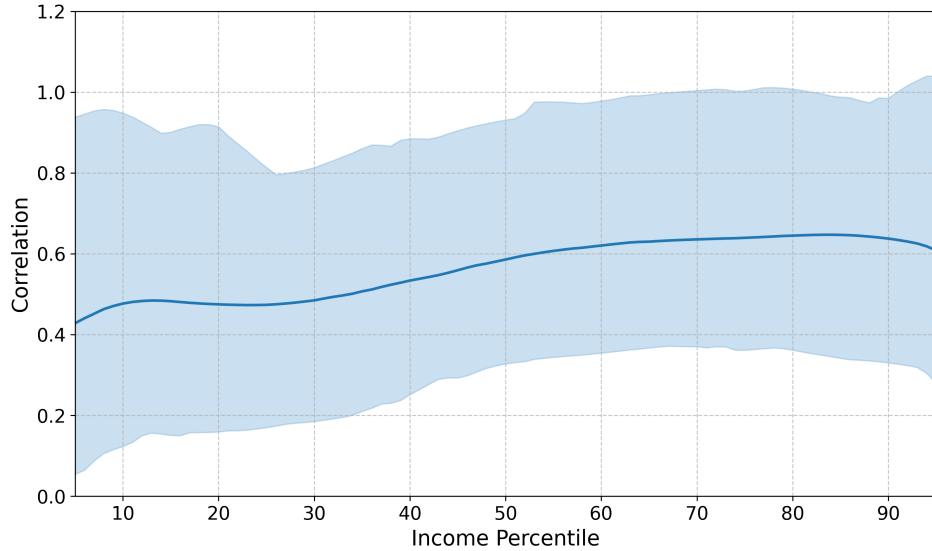


Figure 5: Income-conditional correlation evaluated over income percentiles. Shown with 95% confidence intervals.

Figure 5 shows our estimates of the income-conditional correlation of wage and tax elasticities, by income level. These correlations are consistently positive across incomes and slightly higher at higher incomes. We can statistically reject zero correlation at all income levels. We also form an estimate $\hat{\psi}$ of the average within-income elasticity correlation by averaging across the empirical distribution of year-demeaned incomes. We estimate $\hat{\psi} = 0.56$ with a 95% confidence interval of (0.32, 0.87).

6 Ballpark Quantification

We now bring together the theoretical results from Sections 3 and 4 with our empirical estimates from Section 5. These elements—along with other, more conventional, statistics—allow us to quantify how the cost of redistribution away from top US earners has changed over the last three decades. In effect, we run a “horse race” between the two qualitative mechanisms highlighted by our theory: Decreases in the cost of the redistribution due to *stretching* of the income distribution and increases in the cost of redistribution due to *sorting* of high-elasticity households toward high incomes.

6.1 Changes in the cost of redistribution, 1994 to 2023

To start, recall Lemma 1, which states that changes in the cost of redistribution can be decomposed into changes in the first-order Pareto shape $\alpha(p, \phi)$ and changes in the average within-percentile taxable income elasticity $\varepsilon(p, \phi)$. This result requires that marginal taxes at percentile p are held constant. Conveniently, top tax rates were roughly constant (at 35%) in the US between 1994 and 2023, following the Tax Reform Act of 1993. We therefore focus on changes in the first-order Pareto shape and taxable income elasticities.

We begin with changes in the shape of the income distribution, as captured by $\alpha(p, \phi)$. To the extent $\alpha(p, \phi)$ converges as $p \rightarrow 1$, we are simply interested in measuring the change in the Pareto tail parameter of the income distribution. One simple way to estimate the Pareto tail parameter is, as in Atkinson and Piketty [2010], to use the income shares $s_{1\%}$ and $s_{0.1\%}$ of the top 1% and 0.1% of earners.

$$\alpha_{\text{top}} = \left(1 - \log_{10}(s_{1\%}/s_{0.1\%})\right)^{-1} \quad (38)$$

Data on changes in shares of pre-tax national income from the World Inequality Database show a decline in α_{top} from 1.70 in 1994 to 1.47 in 2023, implying $\Delta \log \alpha_{\text{top}} \approx -0.145$ [Alvaredo et al., 2022].¹⁶

We next turn to changes in average elasticities of taxable income (ETIs) at the top of the distribution, ε_{top} . While some studies, such as Rubolino and Waldenström [2019] and Klemm et al. [2018], have attempted to measure changes in top ETIs over time, their estimates are relatively noisy and face identification challenges. One particular difficulty is that the vast majority of top tax reforms with substantial rate changes took place in the 1980s and 1990s.

Rather than relying on these reduced-form estimates, we take a model-based approach,

¹⁶The distribution for which we would ideally like to observe changes in the Pareto parameter over time is the distribution of adjusted gross income (AGI). The pre-tax national income measure consists of labor and capital income, but does not account for deductions unlike AGI.

asking how top ETIs should have changed given productivity growth between 1994 and 2023. We establish in the proofs of Propositions 4 and 5 that, to first order in uniform productivity changes,

$$\begin{aligned} \Delta \log \varepsilon_{1-T'}(p, \phi) &\approx \mathbb{E} \left[\Delta \log \omega^h \middle| z^h(\phi_0) = \mathcal{Z}(p, \phi_0) \right] \tilde{\alpha}(p, \phi_0) \\ &\times \frac{\mathbb{SD}[\varepsilon_{1-T'}^h \middle| z^h(\phi_0) = \mathcal{Z}(p, \phi_0)] \mathbb{SD}[\varepsilon_w^h \middle| z^h(\phi_0) = \mathcal{Z}(p, \phi_0)]}{\mathbb{E}[\varepsilon_{1-T'}^h \middle| z^h(\phi_0) = \mathcal{Z}(p, \phi_0)]^2} \\ &\times \mathbb{Corr}[\varepsilon_{1-T'}^h, \varepsilon_w^h \middle| z^h(\phi_0) = \mathcal{Z}(p, \phi_0)] \end{aligned} \quad (39)$$

at income percentiles p where taxes are locally linear. We assume productivity changes at top incomes from 1994 to 2023 were uniform and infer them from changes in the income of the 95-99 percentiles of the income distribution. To be conservative, we assume that only half of this group's 33% increase in wage income from 1994 to 2012 is due to productivity improvements, while the rest are elasticity responses to productivity improvements [Bivens et al., 2014].¹⁷ We set $\tilde{\alpha}(p, \phi)$ to 1.70, our estimate of α_{top} from above. For the standard deviations of taxable income elasticities out of tax and wage changes, we apply a conservative lower bound of $\sqrt{0.25}$ based on the estimates of Becko and Sztutman [2024]. For mean ETIs at top incomes, we use the same authors' estimate of the mean ETIs for top earners, 0.3. Finally, our estimates from Section 5 imply a within-income correlation of tax and wage elasticities of 0.6 at top incomes. Combining these estimates implies

$$\Delta \log \varepsilon_{1-T'}^{\text{top}} \approx \log(1.167) \times 1.70 \times 0.25 / (0.3)^2 \times 0.6 = 0.437 \quad (40)$$

Taken together, Lemma 1, observed declines in the Pareto parameter of the income distribution, and model-implied increases in top ETIs imply a substantial rise between 1994 and 2023 in the cost of redistribution away from top incomes. Mathematically,

$$\log \kappa^{\text{top}}(\phi_{2023}) - \log \kappa^{\text{top}}(\phi_{1994}) \approx \underbrace{-0.145}_{\Delta \log \alpha^{\text{top}}} + \underbrace{0.437}_{\Delta \log \varepsilon_{1-T'}^{\text{top}}} = 0.292 \quad (41)$$

On one hand, increases in top income inequality modestly decrease the cost of redistribution by stretching the income distribution. On the other hand, sorting of high-elasticity households to top incomes sharply decrease the cost of redistribution, greatly outweighing the first effect. On net, the cost of redistribution rises by around 34%, from 0.27 to 0.37.¹⁸ That is, in 1994, a tax reform redistributing one dollar to each non-top household had to be accompanied by a lump-sum tax of 27 cents to be budget balanced. Today, it must be accompanied by a lump-sum tax of 0.37 cents.

¹⁷Future versions will account for wage income growth between 2012 and 2023.

¹⁸Here we use the observation that $\kappa(p, \phi) = \alpha(p, \phi)\varepsilon(p, \phi)T'(\mathcal{Z}(p, \phi), \phi)/(1 - T'(\mathcal{Z}(p, \phi), \phi))$.

6.2 Changes in the cost of redistribution, 1979 to 2023

So far, we have focused on changes in the cost of redistribution between 1994 and 2023 because top marginal tax rates are roughly constant over this period. However, this period excludes significant declines in the first-order Pareto shape of the income distribution between 1979 and 1994. The same data sources used above imply that, over this longer period,

$$\Delta \log \alpha^{\text{top}} = -0.246 \quad \text{and} \quad \Delta \log \varepsilon_{1-T'}^{\text{top}} = 0.841 \quad (42)$$

Intuitively, the Pareto shape of the income distribution declines by more over this period due to larger increases in income inequality, but elasticities also increase by more due to larger increases in the wages of top earners. Finally, over this period, top marginal income tax rates also declined from around 50% to 35%; this lowers the cost of redistribution by reducing fiscal externalities.

Combining these effects implies

$$\log \kappa^{\text{top}}(\phi_{2023}) - \log \kappa^{\text{top}}(\phi_{1979}) \approx \underbrace{-0.246}_{\Delta \log \alpha^{\text{top}}} + \underbrace{0.841}_{\Delta \log \varepsilon_{1-T'}^{\text{top}}} + \underbrace{-0.619}_{\Delta \log T'^{\text{top}}/1-T'^{\text{top}}} = 0.024 \quad (43)$$

In other words, were it not for the sorting mechanism, the cost of redistribution would have fallen sharply due to increases in inequality and declines in top taxes between 1979 and 2023. However, the sorting mechanism almost exactly undoes these effects.

7 Conclusion

US income inequality has increased in recent decades. We analyze this phenomenon in light of well-documented heterogeneity in elasticities of labor supply and taxable income, focusing on the fiscal costs of redistributing income.

We define the cost of redistributing income at income percentile p as lump-sum tax required to make up lost revenues from a redistributive reform that increases marginal taxes at the income percentile p to give one dollar to each taxpayer below that income percentile. We provide a decomposition: changes in the costs of redistribution can be driven by either the *stretching* the income distribution or *sorting* if individuals of different elasticities across different income levels.

We can measure *stretching* of the income distribution by changes in its Pareto shape. At the top, these changes account for a 15% decrease in the costs of redistributing income from 1994 to 2023. We provide a procedure to infer the strength of *sorting* mechanism from the joint distribution of income, tax, and wage elasticities. We find that the sorting mechanism has contributed to a 45% increase in the costs of redistributing income.

Together, our estimates imply that the cost of redistribution away from top US earners has risen by 30% since 1994. Since marginal tax rates have been roughly constant, our findings are consistent with an *increased* social value on redistribution in the eyes of policymakers.

References

- Thomas Piketty and Emmanuel Saez. Income inequality in the united states, 1913–1998. *The Quarterly journal of economics*, 118(1):1–41, 2003.
- Thomas Piketty, Emmanuel Saez, and Gabriel Zucman. Distributional national accounts: methods and estimates for the united states. *The Quarterly Journal of Economics*, 133(2):553–609, 2018.
- Gerald Auten and David Splinter. Income inequality in the united states: Using tax data to measure long-term trends. *Journal of Political Economy*, 132(7):000–000, 2024.
- Nada Eissa and Jeffrey B Liebman. Labor supply response to the earned income tax credit. *The quarterly journal of economics*, 111(2):605–637, 1996.
- Jon Gruber and Emmanuel Saez. The elasticity of taxable income: evidence and implications. *Journal of public Economics*, 84(1):1–32, 2002.
- John Sturm Becko and André Sztutman. Income taxation with elasticity heterogeneity. 2024.
- Enrico Rubolino and Daniel Waldenström. Trends and gradients in top tax elasticities: cross-country evidence, 1900–2014. *International Tax and Public Finance*, 26:457–485, 2019.
- Susan Athey, Julie Tibshirani, and Stefan Wager. Generalized random forests. 2019.
- François Bourguignon and Amedeo Spadaro. Tax–benefit revealed social preferences. *The Journal of Economic Inequality*, 10:75–108, 2012.
- Lawrence F Katz and Kevin M Murphy. Changes in relative wages, 1963–1987: supply and demand factors. *The quarterly journal of economics*, 107(1):35–78, 1992.
- David H Autor, Frank Levy, and Richard J Murnane. The skill content of recent technological change: An empirical exploration. *The Quarterly journal of economics*, 118(4): 1279–1333, 2003.
- Daron Acemoglu and Pascual Restrepo. Robots and jobs: Evidence from us labor markets. *Journal of political economy*, 128(6):2188–2244, 2020.

- Daron Acemoglu, David Autor, Jonathon Hazell, and Pascual Restrepo. Artificial intelligence and jobs: Evidence from online vacancies. *Journal of Labor Economics*, 40(S1):S293–S340, 2022.
- Laurence Ales, Musab Kurnaz, and Christopher Sleet. Technical change, wage inequality, and taxes. *American Economic Review*, 105(10):3061–3101, 2015.
- Florian Scheuer and Iván Werning. The taxation of superstars. *The Quarterly Journal of Economics*, 132(1):211–270, 2017.
- Jonathan Heathcote, Kjetil Storesletten, and Giovanni L Violante. Optimal tax progressivity: An analytical framework. *The Quarterly Journal of Economics*, 132(4):1693–1754, 2017.
- Chunzan Wu. More unequal income but less progressive taxation. *Journal of Monetary Economics*, 117:949–968, 2021.
- Vidar Christiansen and Eilev S Jansen. Implicit social preferences in the norwegian system of indirect taxation. *Journal of Public Economics*, 10(2):217–245, 1978.
- Nathaniel Hendren. Measuring economic efficiency using inverse-optimum weights. *Journal of public Economics*, 187:104198, 2020.
- Nathaniel Hendren and Ben Sprung-Keyser. A unified welfare analysis of government policies. *The Quarterly Journal of Economics*, 135(3):1209–1318, 2020.
- Florian Scheuer and Iván Werning. Mirrlees meets diamond-mirrlees. Technical report, National Bureau of Economic Research, 2016.
- Laurence Jacquet and Etienne Lehmann. Optimal income taxation with composition effects. *Journal of the European Economic Association*, 19(2):1299–1341, 2021.
- Felix J Bierbrauer, Pierre C Boyer, and Emanuel Hansen. Pareto-improving tax reforms and the earned income tax credit. *Econometrica*, 91(3):1077–1103, 2023.
- Martin Feldstein. Tax avoidance and the deadweight loss of the income tax. *Review of Economics and Statistics*, 81(4):674–680, 1999.
- Joel Slemrod. Income creation or income shifting? behavioral responses to the tax reform act of 1986. *The American Economic Review*, 85(2):175–180, 1995.
- Joel Slemrod. Tax avoidance, evasion, and administration. *Handbook of Public Economics*, 2002.

- Raj Chetty. Is the taxable income elasticity sufficient to calculate deadweight loss? the implications of evasion and avoidance. *American Economic Journal: Economic Policy*, 1(2):31–52, 2009.
- Francine D Blau and Lawrence M Kahn. Changes in the labor supply behavior of married women: 1980–2000. *Journal of Labor economics*, 25(3):393–438, 2007.
- Anil Kumar and Che-Yuan Liang. Estimating taxable income responses with elasticity heterogeneity. *Journal of Public Economics*, 188:104209, 2020.
- Emmanuel Saez. Using elasticities to derive optimal income tax rates. *The review of economic studies*, 68(1):205–229, 2001.
- Florian Scheuer and Iván Werning. Mirrlees meets diamond-mirrlees: simplifying non-linear income taxation. *NBER Working papers*, (22076), 2018.
- Bas Jacobs, Egbert LW Jongen, and Floris T Zoutman. Revealed social preferences of dutch political parties. *Journal of Public Economics*, 156:81–100, 2017.
- Daniel Feenberg and Elisabeth Coutts. An introduction to the taxsim model. *Journal of Policy Analysis and Management*, 12(1):189–194, 1993. ISSN 02768739, 15206688. URL <http://www.jstor.org/stable/3325474>.
- A.B. Atkinson and T. Piketty. *Top Incomes: A Global Perspective*. OUP Oxford, 2010. ISBN 9780199286898. URL <https://books.google.com/books?id=zLQUDAAAQBAJ>.
- Facundo Alvaredo, Anthony B. Atkinson, Thomas Piketty, and Emmanuel Saez. World inequality database, 2022. URL <http://wid.world/data>.
- Mr Alexander D Klemm, Ms Li Liu, Victor Mylonas, and Mr Philippe Wingender. Are elasticities of taxable income rising? *IMF Working Paper*, 2018.
- Josh Bivens, Elise Gould, E Mishel, and Heidi Shierholz. Raising america’s pay. *Economic Policy Institute Briefing Paper*, 378, 2014.

Appendix

A Proofs

For technical convenience, all of our proofs work with income retention schedules that map pre-tax income to post-tax income rather than tax schedules $T^h(\cdot, \phi)$ that map pre-tax income to taxes owed. The retention schedules $R(\cdot, \phi)$ that corresponds to the tax schedules $T^h(\cdot, \phi)$ used in the main text are defined by

$$R^h(z, \phi) = z - T^h(z, \phi) \quad (44)$$

A.1 First order changes in household behavior

We now derive an expression for first-order changes in household behavior with respect to changes in the income retention schedule and technology.

Recall the household problem at an arbitrary retention schedule R and technology ϕ :

$$\max_{\ell \geq 0} u^h(R(\ell w^h(\phi)), \ell) \quad (45)$$

Let $\ell^h(R, \phi)$ denote this problem's solution. Assuming $\ell^h(R, \phi) > 0$ and u^h and R are differentiable, we have the first-order condition

$$0 = R'(w^h(\phi)\ell^h(R, \phi))w^h(\phi) - M^h(R(w^h(\phi)\ell^h(R, \phi)), \ell^h(R, \phi)) \quad (46)$$

where $M^h(c, \ell) \equiv -u_\ell^h(c, \ell)/u_c^h(c, \ell)$.

We now apply the implicit function theorem to (46). To simplify expressions we let $z^h(R, \phi) \equiv w^h(\phi)\ell^h(R, \phi)$, $c^h(R, \phi) = R(z^h(R, \phi))$, $M_c^h(R, \phi) \equiv M_c^h(c^h(R, \phi), \ell^h(R, \phi))$, $M_\ell^h(R, \phi) \equiv M_\ell^h(c^h(R, \phi), \ell^h(R, \phi))$.

$$\begin{aligned} & D\ell^h(R, \phi) \cdot (\Delta, 1) \\ &= - \frac{\Delta'(z^h(R, \phi))w^h(\phi) - M_c^h(R, \phi)\Delta(z^h(R, \phi))}{R''(z^h(R, \phi))w^h(\phi)^2 - M_c^h(R, \phi)R'(z^h(R, \phi))w^h(\phi) - M_\ell^h(R, \phi)} \\ &\quad - \frac{R''(z^h(R, \phi))z^h(R, \phi) + R'(z^h(R, \phi)) - M_c^h(R, \phi)R'(z^h(R, \phi))\ell^h(R, \phi)}{R''(z^h(R, \phi))w^h(\phi)^2 - M_c^h(R, \phi)R'(z^h(R, \phi))w^h(\phi) - M_\ell^h(R, \phi)} w^{h'}(\phi) \\ &= - \frac{\frac{\Delta'(z^h(R, \phi))}{R'(z^h(R, \phi))} - \frac{M_c^h(R, \phi)c^h(R, \phi)}{M^h(R, \phi)} \frac{\Delta(z^h(R, \phi))}{R(z^h(R, \phi))}}{\frac{R''(z^h(R, \phi))z^h(R, \phi)}{R'(z^h(R, \phi))} \frac{1}{\ell^h(R, \phi)} - \frac{M_c^h(R, \phi)c^h(R, \phi)}{M^h(R, \phi)} \frac{R'(z^h(R, \phi))z^h(R, \phi)}{R(z^h(R, \phi))} \frac{1}{\ell^h(R, \phi)} - \frac{M_\ell^h(R, \phi)\ell^h(R, \phi)}{M^h(R, \phi)} \frac{1}{\ell^h(R, \phi)}} \end{aligned}$$

$$\begin{aligned}
& - \frac{\frac{R''(z^h(R, \phi))z^h(R, \phi)}{R'(z^h(R, \phi))} + 1 - \frac{M_c^h(R, \phi)c^h(R, \phi)}{M^h(R, \phi)} \frac{R'(z^h(R, \phi))z^h(R, \phi)}{R(z^h(R, \phi))}}{\frac{R''(z^h(R, \phi))z^h(R, \phi)}{R'(z^h(R, \phi))} \frac{1}{\ell^h(R, \phi)} - \frac{M_c^h(R, \phi)c^h(R, \phi)}{M^h(R, \phi)} \frac{R'(z^h(R, \phi))z^h(R, \phi)}{R(z^h(R, \phi))} \frac{1}{\ell^h(R, \phi)} - \frac{M_\ell^h(R, \phi)\ell^h(R, \phi)}{M^h(R, \phi)} \frac{1}{\ell^h(R, \phi)}} \frac{w^{h'}(\phi)}{w^h(\phi)} \\
& \quad (47)
\end{aligned}$$

Rearranging, we have

$$\begin{aligned}
& D \log \ell^h(R, \phi) \cdot (\Delta, 1) \\
& = \frac{\frac{\Delta'(z^h)}{R'(z^h)} - \frac{d \log M_c^h(c^h, \ell^h)}{d \log c^h} \frac{\Delta(z^h)}{R(z^h)}}{\frac{d \log M_c^h(c^h, \ell^h)}{d \log c^h} \frac{d \log R(z^h)}{d \log z^h} + \frac{d \log M_\ell^h(c^h, \ell^h)}{d \log \ell^h} - \frac{d \log R'(z^h)}{d \log z^h}} \Bigg|_{\substack{z^h=z^h(R, \phi) \\ c^h=c^h(R, \phi) \\ \ell^h=\ell^h(R, \phi)}} \\
& + \frac{\frac{d \log R'(z^h)}{d \log z^h} + 1 - \frac{d \log M_c^h(c^h, \ell^h)}{d \log c^h} \frac{d \log R(z^h)}{d \log z^h}}{\frac{d \log M_c^h(c^h, \ell^h)}{d \log c^h} \frac{d \log R(z^h)}{d \log z^h} + \frac{d \log M_\ell^h(c^h, \ell^h)}{d \log \ell^h} - \frac{d \log R'(z^h)}{d \log z^h}} \Bigg|_{\substack{z^h=z^h(R, \phi) \\ c^h=c^h(R, \phi) \\ \ell^h=\ell^h(R, \phi)}} \frac{d \log w^h(\phi)}{d \phi} \\
& \quad (48)
\end{aligned}$$

We conclude that

$$\begin{aligned}
D \log \ell^h(R, \phi) \cdot (\Delta, 1) &= \varepsilon^h(R, \phi) \left[\frac{\Delta'(z^h)}{R'(z^h)} + \left(1 + \frac{d \log R'(z^h)}{d \log z^h} \right) \frac{d \log w^h(\phi)}{d \phi} \right] \\
&+ \eta^h(R, \phi) \left[\frac{\Delta(z^h)}{R(z^h)} + \frac{d \log R(z^h)}{d \log z^h} \frac{d \log w^h(\phi)}{d \phi} \right] \Bigg|_{z^h=z^h(R, \phi)} \\
\text{where } \varepsilon^h(R, \phi) &\equiv \frac{1}{\frac{d \log M_c^h(c^h, \ell^h)}{d \log c^h} \frac{d \log R(z^h)}{d \log z^h} + \frac{d \log M_\ell^h(c^h, \ell^h)}{d \log \ell^h} - \frac{d \log R'(z^h)}{d \log z^h}} \Bigg|_{\substack{z^h=z^h(R, \phi) \\ c^h=c^h(R, \phi) \\ \ell^h=\ell^h(R, \phi)}} \\
\text{and } \eta^h(R, \phi) &\equiv \frac{-\frac{d \log M_c^h(c^h, \ell^h)}{d \log c^h}}{\frac{d \log M_c^h(c^h, \ell^h)}{d \log c^h} \frac{d \log R(z^h)}{d \log z^h} + \frac{d \log M_\ell^h(c^h, \ell^h)}{d \log \ell^h} - \frac{d \log R'(z^h)}{d \log z^h}} \Bigg|_{\substack{z^h=z^h(R, \phi) \\ c^h=c^h(R, \phi) \\ \ell^h=\ell^h(R, \phi)}} \\
& \quad (49)
\end{aligned}$$

It follows that

$$\begin{aligned}
D \log z^h(R, \phi) \cdot (\Delta, 1) &= \frac{d \log w^h(\phi)}{d \phi} \\
&+ \varepsilon^h(R, \phi) \left[\frac{\Delta'(z^h)}{R'(z^h)} + \left(1 + \frac{d \log R'(z^h)}{d \log z^h} \right) \frac{d \log w^h(\phi)}{d \phi} \right] \\
&+ \eta^h(R, \phi) \left[\frac{\Delta(z^h)}{R(z^h)} + \frac{d \log R(z^h)}{d \log z^h} \frac{d \log w^h(\phi)}{d \phi} \right] \Bigg|_{z^h=z^h(R, \phi)} \\
& \quad (50)
\end{aligned}$$

Moreover, recalling $z^h(\phi) \equiv z^h(R(\phi), \phi)$ and similarly defining $\varepsilon(\phi) \equiv \varepsilon(R(\phi), \phi)$ and

$\eta(\phi) \equiv \eta(R(\phi), \phi)$, we have

$$\begin{aligned} \frac{d \log z^h(\phi)}{d\phi} &= \left(1 + \varepsilon^h(\phi)\right) \frac{d \log w^h(\phi)}{d\phi} \\ &+ \varepsilon^h(\phi) \frac{d \log R'(w^h(\phi)\ell^h; \phi)}{d\phi} \Big|_{\ell^h=\ell^h(\phi)} \\ &+ \eta^h(\phi) \frac{d \log R(w^h(\phi)\ell^h; \phi)}{d\phi} \Big|_{\ell^h=\ell^h(\phi)} \end{aligned} \quad (51)$$

A.2 Second-order effects on behavior

Fix some p and δp . Suppressing them from our notation for legibility, define

$$z^h(\epsilon, \phi) \equiv z^h(R(\epsilon, \phi), \phi), \quad R(\epsilon, \phi) \equiv R(\phi) + \epsilon \Delta(p, \delta p; \phi). \quad (52)$$

From the previous section, we have

$$\begin{aligned} \log z_\phi^h(\epsilon, \phi) &= \frac{d \log w^h(\phi)}{d\phi} \\ &+ \varepsilon^h(\epsilon, \phi) \left[\frac{R'_\phi(z^h; \epsilon, \phi)}{R(z^h; \epsilon, \phi)} + \left(1 + \frac{d \log R'(z^h; \epsilon, \phi)}{d \log z^h}\right) \frac{d \log w^h(\phi)}{d\phi} \right] \\ &+ \eta^h(\epsilon, \phi) \left[\frac{R_\phi(z^h; \epsilon, \phi)}{R(z^h; \epsilon, \phi)} + \frac{d \log R(z^h; \epsilon, \phi)}{d \log z^h} \frac{d \log w^h(\phi)}{d\phi} \right] \Big|_{z^h=z^h(\epsilon, \phi)} \\ &= \left(1 + \varepsilon^h(\epsilon, \phi)\right) \frac{d \log w^h(\phi)}{d\phi} \\ &+ \varepsilon^h(\epsilon, \phi) \frac{d \log R'(w^h(\phi)\ell^h; \epsilon, \phi)}{d\phi} \Big|_{\ell^h=\ell^h(\epsilon, \phi)} \\ &+ \eta^h(\epsilon, \phi) \frac{d \log R(w^h(\phi)\ell^h; \epsilon, \phi)}{d\phi} \Big|_{\ell^h=\ell^h(\epsilon, \phi)} \\ &= \left(1 + \varepsilon^h(\epsilon, \phi)\right) \frac{d \log w^h(\phi)}{d\phi} \\ &+ \varepsilon^h(\epsilon, \phi) \frac{d \log R'(w^h(\phi)\ell^h; \epsilon, \phi)}{d\phi} \Big|_{\ell^h=\ell^h(\epsilon, \phi)} \\ &+ \eta^h(\epsilon, \phi) \frac{d \log R(w^h(\phi)\ell^h; \epsilon, \phi)}{d\phi} \Big|_{\ell^h=\ell^h(\epsilon, \phi)} \end{aligned} \quad (53)$$

Now consider the second derivative with respect to ϵ :

$$\begin{aligned} \log z_{\phi\epsilon}^h(\epsilon, \phi) &= \left[\varepsilon_\epsilon^h(\epsilon, \phi) \left(\frac{d \log w^h(\phi)}{d\phi} + \frac{d \log R'(w^h(\phi)\ell^h; \epsilon, \phi)}{d\phi} \right) + \eta_\epsilon^h(\epsilon, \phi) \frac{d \log R(w^h(\phi)\ell^h; \epsilon, \phi)}{d\phi} \right] \Big|_{\ell^h=\ell^h(\epsilon, \phi)} \\ &+ \left[\varepsilon^h(\epsilon, \phi) \frac{d^2 \log R'(w^h(\phi)\ell^h; \epsilon, \phi)}{d\phi d\epsilon} + \eta^h(\epsilon, \phi) \frac{d^2 \log R(w^h(\phi)\ell^h; \epsilon, \phi)}{d\phi d\epsilon} \right] \Big|_{\ell^h=\ell^h(\epsilon, \phi)} \end{aligned} \quad (54)$$

A.3 Cost of Redistribution

We begin by deriving the cost of redistribution with general household utility. We use the fact (see [Becko and Sztutman \[2024\]](#) for a derivation) that, provided income is supplied on the intensive margin and there are no jumps,

$$\begin{aligned}
D \log z^h(R; \phi) \cdot \Delta &= \varepsilon^h(R; \phi) \frac{\Delta'(z)}{R'(z)} \Big|_{z=z^h(R; \phi)} + \eta^h(R; \phi) \frac{\Delta(z)}{R(z)} \Big|_{z=z^h(R; \phi)} \\
\text{where } \varepsilon^h(R; \phi) &\equiv \frac{1}{\frac{d \log M^h(c^h, z^h; \phi)}{d \log c^h} \frac{d \log R(z^h)}{d \log z^h} + \frac{d \log M^h(c^h, z^h; \phi)}{d \log z^h} - \frac{d \log R'(z^h)}{d \log z^h}} \Big|_{\substack{z^h=z^h(R; \phi), \\ c^h=R(z^h)}} \\
\eta^h(R; \phi) &\equiv \frac{-\frac{d \log M^h(c^h, z^h; \phi)}{d \log c^h}}{\frac{d \log M^h(c^h, z^h; \phi)}{d \log c^h} \frac{d \log R(z^h)}{d \log z^h} + \frac{d \log M^h(c^h, z^h; \phi)}{d \log z^h} - \frac{d \log R'(z^h)}{d \log z^h}} \Big|_{\substack{z^h=z^h(R; \phi), \\ c^h=R(z^h)}} \\
M^h(c, z; \phi) &\equiv -\frac{u_z^h(c, z/w^h(\phi))}{u_c^h(c, z/w^h(\phi))}
\end{aligned} \tag{55}$$

Let $\Delta(\cdot, \delta p) \equiv \Delta(\cdot, p, \delta p, \phi)$ and $R(\cdot) = R(\cdot, \phi)$. We can now calculate the fiscal cost of a redistributive tax reform, denoted as $\zeta(p, \phi)$.

$$\begin{aligned}
\zeta(p, \phi) &= \lim_{\delta p \rightarrow 0} \Big|_{\delta p=0} \frac{d}{d\epsilon} \Big|_{\epsilon=0} \text{Rev}(R + \epsilon \Delta(\delta p), \phi) \\
&= \lim_{\delta p \rightarrow 0} \Big|_{\delta p=0} \mathbb{E} \left[(1 - R'(z^h(\phi))) z^h(\phi) D \log z^h(R; \phi) \cdot \Delta(\delta p) - \underbrace{\Delta(z^h(\phi), \delta p)}_{\text{no mech. eff.}} \right] \\
&= \lim_{\delta p \rightarrow 0} \Big|_{\delta p=0} \mathbb{E} \left[(1 - R'(z^h(\phi))) z^h(\phi) \varepsilon^h(R; \phi) \frac{\Delta'(z^h(\phi), \delta p)}{R'(z^h(\phi))} \right] \\
&\quad - \mathbb{E} \left[(1 - R'(z^h(\phi))) z^h(\phi) \eta^h(R; \phi) \frac{\Delta(z^h(\phi), \delta p)}{R(z^h(\phi))} \right] \\
&= \lim_{\delta p \rightarrow 0} \Big|_{\delta p=0} \mathbb{E} \left[\frac{1 - R'(z^h(\phi))}{R'(z^h(\phi))} z^h(\phi) \varepsilon^h(R; \phi) \right. \\
&\quad \times \frac{(1 + \frac{p}{1-p}) \delta p}{\mathcal{Z}(p + \delta p, \phi) - \mathcal{Z}(p, \phi)} \Big| \mathcal{Z}(p, \phi) < z^h(\phi) \leq \mathcal{Z}(p + \delta p, \phi) \Big] \\
&\quad - \mathbb{E} \left[\frac{1 - R'(z^h(\phi))}{R(z^h(\phi))} z^h(\phi) \eta^h(R; \phi) \Big| z^h(\phi) \leq \mathcal{Z}(p, \phi) \right] p \\
&\quad + \mathbb{E} \left[\frac{1 - R'(z^h(\phi))}{R(z^h(\phi))} z^h(\phi) \eta^h(R; \phi) \Big| z^h(\phi) > \mathcal{Z}(p, \phi) \right] (1 - p)
\end{aligned} \tag{56}$$

Denoting the last two terms by $\Gamma(p, \phi)$, recalling that $\mathcal{Z}(p, \phi)$ is differentiable, and

using the fact that $\mathcal{Z}_p(p, \phi) = f(z, \phi)^{-1}$ by the inverse function theorem, we have

$$\zeta(p, \phi) = \frac{1 - R'(z)}{R'(z)} \underbrace{\mathbb{E}[\varepsilon^h(\phi) \mid z^h(\phi) = z]}_{\varepsilon(p, \phi)} \underbrace{\frac{zf(z, \phi)}{1 - p}}_{\equiv \alpha(p, \phi)} + \Gamma(p, \phi) \Big|_{z=\mathcal{Z}(p, \phi)} \quad (57)$$

We conclude that

$$\begin{aligned} \zeta(p, \phi) &= \frac{1 - R'(z; \phi)}{R'(z; \phi)} \alpha(z; \phi) \varepsilon(z; \phi) \\ &\quad - \int_0^z \frac{1 - R'(\tilde{z}; \phi)}{R'(\tilde{z}; \phi)} \frac{d \log R'(\tilde{z}; \phi)}{d \log \tilde{z}} \eta(\tilde{z}; \phi) d\tilde{z} \\ &\quad + \int_z^\infty \frac{1 - R'(\tilde{z}; \phi)}{R'(\tilde{z}; \phi)} \frac{d \log R'(\tilde{z}; \phi)}{d \log \tilde{z}} \eta(\tilde{z}; \phi) d\tilde{z} \quad \text{for } z = \mathcal{Z}(p, \phi) \end{aligned} \quad (58)$$

where $\eta(\tilde{z}; \phi) \equiv \mathbb{E}[\eta^h(R; \phi) \mid z^h(\phi) = \tilde{z}]$ are income-conditional mean elasticities at $\tilde{z}$ and technology ϕ .

Lemma A.1. *Assuming Assumption 2 holds, which assumes preferences are quasilinear in the numeraire, then $\forall p \in (0, 1)$ and technologies ϕ and ϕ' ,*

$$\log \kappa(p, \phi') - \log \kappa(p, \phi) = \left[\log \alpha(p, \phi') - \log \alpha(p, \phi) \right] + \left[\log \varepsilon(p, \phi') - \log \varepsilon(p, \phi) \right] \quad (59)$$

Proof. As argued in the main text, when labor supply does not respond to lump-sum transfers, $\kappa(p, \phi) = \zeta(p, \phi)$. Moreover, $\Gamma(p, \phi) = 0$ in this case. So with Assumption 2,

$$\kappa(p, \phi) = \zeta(p, \phi) = \frac{1 - R'(z)}{R'(z)} \underbrace{\mathbb{E}[\varepsilon^h(\phi) \mid z^h(\phi) = z]}_{\varepsilon(p, \phi)} \underbrace{\frac{zf(z, \phi)}{1 - p}}_{\equiv \alpha(p, \phi)} \Big|_{z=\mathcal{Z}(p, \phi)} \quad (60)$$

The decomposition follows by taking logs of $\kappa(p, \phi)$ and $\kappa(p, \phi')$ and differencing. $\square$

B Derivations of main propositions and supporting lemmas

B.1 Technical Conditions and the Construction Lemma

In this section, we will introduce a lemma that constructs a tax schedule under which household optimization ensures marginal rate at every percentile is the same as the baseline. This lemma applies for Propositions 1 through 3, which assumes homogeneity in β^h , and will be frequently invoked in the proofs of these propositions. Let us first define a few useful pieces of notation for the lemma.

Definition 1. $(\hat{z}, \hat{F}, \hat{Z})$: Let $r^h(\phi_0)$ denote the marginal rate h faces at the baseline,

$$\hat{z}^h(\phi) \equiv r^h(\phi_0)^{\beta^h} w^h(\phi)^{1+\beta^h} \quad (61)$$

$\hat{F}(\cdot, \phi)$ denotes the resulting income CDF, $\hat{f}(\cdot, \phi)$ its PDF, and $\hat{Z} \equiv \hat{F}^{-1}$.

Definition 2. The tax schedule $\hat{R}(\cdot, \phi)$ is defined as

$$\hat{R}(z, \phi) \equiv \int_0^{\hat{F}(z, \phi)} \frac{r(\mathcal{Z}(p, \phi_0), \phi_0)}{\hat{f}(\hat{Z}(p, \phi), \phi)} dp \quad (62)$$

We also introduce some technical conditions below for our propositions; only a subset of which will be necessary for each result.

Assumption 3. For any $\phi \in \Phi$, $\phi \neq \phi_0$, $\exists! \tilde{R}(\cdot)$ up to a constant so that given $(\phi, \tilde{R}(\cdot))$, the following expression holds $\forall p \in (0, 1)$

$$\tilde{R}(\tilde{Z}(p, \phi)) = R(\mathcal{Z}(p, \phi_0), \phi_0) \quad (63)$$

Where $\tilde{Z}(p, \phi_1)$ is the inverse CDF of the income distribution arising from household optimization given $(\phi, \tilde{R}(\cdot))$.

Assumption 4. For a given $\phi \in \Phi$, labor is supplied on the intensive margin with no jumps when households optimize facing $(\hat{R}(\cdot, \phi), \phi)$.

Assumption 5. (Regularity of productivity change across households)

For some $0 < L < \infty$, $\exists \delta > 0$ such that for all $\phi \in [\phi_0, \phi_0 + \delta)$, we have

$$|(\tilde{w}^{h'}(\phi) - \tilde{w}^{h'}(\phi_0)) - (\tilde{w}^h(\phi) - \tilde{w}^h(\phi_0))| \leq L |\tilde{z}^h(\phi_0) - \tilde{z}^{h'}(\phi_0)| |\phi - \phi_0| \quad \forall h, h' \in \mathcal{H} \quad (64)$$

Assumption 6. (Regularity of across income variation in β)

Let $\beta(z) = \mathbb{E}[\beta^h | z^h(\phi_0) = z]$. β is differentiable and for some $0 < L < \infty$,

$$|\beta(z) - \beta(z')| \leq L |\log z - \log z'| \quad \forall z, z' \in (0, \infty) \quad (65)$$

We are now ready to introduce Lemma B.2 that constructs the unique tax schedule under the assumptions made in any of the first three propositions. The structure of the proof is as follows. We take a “guess and verify” approach by first assuming HHs set their new income to $\hat{z}^h(\phi)$ as defined above. We then verify that the resulting retention schedule from $\hat{R}$ ensures that FOCs for all households are satisfied when facing $(\hat{R}(\cdot, \phi), \phi)$. Since our choice of projection holds marginal retention at every percentile fixed, if the relative order of HHs as ordered by their baseline incomes do not change before and after the shock, their new marginal retention would be the same as before, so $\hat{z}^h(\phi)$ would coincide with their optimizing income $z^h(\phi)$. Lemma B.1 formally verifies that the order does not change under assumptions in Propositions 1, 2, or 3.

Lemma B.1. *Under the set of conditions in either Proposition 1, 2, or 3, $\forall h, h' \in \mathcal{H}$*

$$z^h(\phi_0) > z^{h'}(\phi_0) \implies \hat{z}^h(\phi) > \hat{z}^{h'}(\phi) \quad (66)$$

$\forall \phi \in \Phi$ in Proposition 1 and $\forall \phi \in B_\delta(\phi_0)$ with $\delta > 0$ suitably small in Proposition 2 and Proposition 3.

Proof. We prove the lemma for each proposition individually using only assumptions specified in that proposition.

1. **Proposition 1:** As we will argue in the neutral benchmark, $\hat{z}^h(\phi) = cz^h(\phi_0) \forall h$, where $c > 0$ is the same constant for all households, so the no swapping condition follows immediately.
2. **Proposition 2:** The baseline FOC is

$$\tilde{z}^h(\phi_0) = \beta \tilde{r}^h(\phi_0) + (1 + \beta) \tilde{w}^h(\phi_0) \quad (67)$$

where $\tilde{x} = \log(x)$. Note

$$\begin{aligned} \tilde{\hat{z}}^h(\phi) - \tilde{\hat{z}}^{h'}(\phi) &= (\tilde{\hat{z}}^h(\phi) - \tilde{z}^h(\phi_0)) - (\tilde{\hat{z}}^{h'}(\phi) - \tilde{z}^{h'}(\phi_0)) + (\tilde{z}^h(\phi_0) - \tilde{z}^{h'}(\phi_0)) \\ &= (1 + \beta)(\tilde{w}^h(\phi) - \tilde{w}^h(\phi_0)) - (1 + \beta)(\tilde{w}^{h'}(\phi) - \tilde{w}^{h'}(\phi_0)) \\ &\quad + (\tilde{z}^h(\phi_0) - \tilde{z}^{h'}(\phi_0)) \end{aligned} \quad (68)$$

Therefore it is equivalent to show that there exists $\delta > 0$ so that for any $\phi \in B_\delta(\phi_0)$,

$$\tilde{z}^h(\phi_0) - \tilde{z}^{h'}(\phi_0) \geq (1 + \beta)(\tilde{w}^{h'}(\phi) - \tilde{w}^{h'}(\phi_0)) - (1 + \beta)(\tilde{w}^h(\phi) - \tilde{w}^h(\phi_0)) \quad (69)$$

From Assumption (5), which we assume in Proposition 2, the RHS can be bounded by $2(1 + \beta)L|\tilde{z}^h(\phi_0) - \tilde{z}^{h'}(\phi_0)||\phi - \phi_0|$ for $\phi \in B_\delta(\phi_0)$. Dividing through by LHS, which is positive by assumption, the claim holds by choosing $\delta < \frac{1}{2(1+\beta)L}$.

3. **Proposition 3:** Proposition 3 assumes Assumption (1), which ensures

$$\tilde{w}^{h'}(\phi) - \tilde{w}^{h'}(\phi_0) = \tilde{w}^h(\phi) - \tilde{w}^h(\phi_0) \equiv c(\phi, \phi_0) \quad \forall h, h' \in \Phi \quad (70)$$

where $c(\phi, \phi_0)$ is continuous in $\phi - \phi_0$ with $c(\phi_0, \phi_0) = 0$. Following the same steps as (68), our claim is equivalent to showing there exists some $\delta > 0$ so that for any $\phi \in B_\delta(\phi_0)$,

$$\tilde{z}^h(\phi_0) - \tilde{z}^{h'}(\phi_0) \geq (\beta^{h'} - \beta^h)c(\phi, \phi_0) \quad (71)$$

The RHS can be written as $(\beta(\tilde{z}^{h'}(\phi_0)) - \beta(\tilde{z}^h(\phi_0)))c(\phi, \phi_0)$ by definition of $\beta(\cdot)$ and Assumption (3) which we assume in Proposition 3. By Assumption (6), the RHS can be bounded from above by $L(\tilde{z}^h(\phi_0) - \tilde{z}^{h'}(\phi_0))c(\phi, \phi_0)$. Dividing both sides through by $(\tilde{z}^h(\phi_0) - \tilde{z}^{h'}(\phi_0))$, the claim follows as for some $\delta > 0$, $c(\phi, \phi_0)L < 1 \forall \phi \in B_\delta(\phi_0)$ by continuity. □

Lemma B.2. (*Construction Lemma*): Under the set of conditions in either Proposition 1, 2, or 3,

$$\hat{R}(\cdot, \phi) \equiv \int_0^{\hat{F}(\cdot, \phi)} \frac{r(\mathcal{Z}(p, \phi_0), \phi_0)}{\hat{f}(\hat{\mathcal{Z}}(p, \phi), \phi)} dp \quad (72)$$

is a schedule satisfying our projection requirement (10), which holds the marginal retention at every percentile fixed, $\forall \phi \in \Phi$ in Proposition 1 and $\forall \phi \in B_\delta(\phi_0)$ for $\delta > 0$ suitably small in Propositions 2 and 3.

Proof. Differentiating $\hat{R}(z, \phi)$ w.r.t. z , we have

$$\hat{r}(z, \phi) = \hat{r}(\hat{\mathcal{Z}}(\hat{F}(z, \phi), \phi), \phi) = r(\mathcal{Z}(\hat{F}(z, \phi), \phi_0), \phi_0) \quad \forall z \quad (73)$$

The projection requirement (10) is shown as long as we can verify that $\hat{z}^h(\phi)$ satisfies the actual FOC. Suppose $z^h(\phi_0) > z^{h'}(\phi_0) \implies \hat{z}^h(\phi) > \hat{z}^{h'}(\phi)$, then $\hat{F}(\hat{z}^h(\phi), \phi) = F(z^h(\phi_0), \phi_0)$, which means $r^h(\phi_0) = \hat{r}(\hat{z}^h(\phi), \phi)$. Finally, the sufficiency of FOC is given by Assumption (4), which we make in all of our propositions, so we are done.

By Lemma B.1, $z^h(\phi_0) > z^{h'}(\phi_0) \implies \hat{z}^h(\phi) > \hat{z}^{h'}(\phi)$, under conditions in each of our first three propositions, so we are done. □

We remark that the uniqueness of a tax schedule at ϕ satisfying our projection requirement (10) is assumed in all propositions to avoid indeterminacy in the tax schedule and therefore indeterminacy in the income distribution. That said, the assumption is separate from the existence proof supplied above. The reader can always assume that in the background we are choosing $\hat{R}(\cdot, \phi)$ consistently if the uniqueness assumption is irksome. Finally, we will frequently make use of the fact that households maintain the

same relative order as sorted by their baseline income at the new ϕ . We summarize this fact in Corollary B.2.1.

Corollary B.2.1. *(No Swapping): Under the set of conditions in either Proposition 1, 2, or 3, we have*

1. $\hat{z}^h(\phi) = z^h(\phi) \forall h$
2. $z^h(\phi_0) > z^{h'}(\phi_0) \implies z^h(\phi) > z^{h'}(\phi) \forall h, h'$

These results hold $\forall \phi \in \Phi$ in Proposition 1 and for any $\forall \phi \in B_\delta(\phi_0)$ for $\delta > 0$ suitably small in Proposition 2 and Proposition 3.

Proof. All of our propositions maintain Assumption (3), so by Lemma B.2, the retention schedule at ϕ is $\hat{R}(\cdot, \phi)$; moreover, all propositions maintain Assumption (4) as well, so if we find an income satisfying the HH FOC, it is the unique optimizing solution.

Under $\hat{R}(\cdot, \phi)$, $\hat{z}^h(\phi)$ solves the HH FOC, so it is the optimizing choice. Having shown $\hat{z}^h(\phi) = z^h(\phi)$, the second statement follows immediately from the statement of Lemma B.1. □

B.2 Propositions

B.2.1 A Neutral Benchmark

Proposition. *If Conditions 1–3 hold and technical assumptions (3) and (4) hold, then for all p and all ϕ_1 , $\kappa(p, \phi_0) = \kappa(p, \phi_1)$.*

Proof. From (57), we have

$$\kappa(p, \phi) = \frac{1 - R'(\mathcal{Z}(p, \phi))}{R'(z, \phi)} \frac{\mathcal{Z}(p, \phi)f(z, \phi)}{1 - p} \mathbb{E}[\varepsilon^h(\phi) \mid z^h(\phi) = \mathcal{Z}(p, \phi)] \quad (74)$$

Under Condition 1, $\left(\frac{w^h(\phi_1)}{w^h(\phi_0)}\right)^{1+\beta} > 0$ is a constant, which we will write as c . Since $\hat{z}^h(\phi) = cz^h(\phi_0)$, we must have $z^h(\phi_0) > z^{h'}(\phi_0) \implies \hat{z}^h(\phi_1) > \hat{z}^{h'}(\phi_1) \forall h, h'$. As such, Lemma B.1 and thereby Lemma B.2 holds. Immediately from Corollary B.2.1:

$$\mathcal{Z}(p, \phi) = \mathcal{Z}(p, \phi_0)c \quad f(\mathcal{Z}(p, \phi_1), \phi_1) = \frac{f(\mathcal{Z}(p, \phi_0), \phi_0)}{c} \quad (75)$$

Lastly, we consider changes to the elasticity term

$$\mathbb{E}[\varepsilon^h(\phi) \mid z^h(\phi) = \mathcal{Z}(p, \phi)] = \frac{\beta}{1 - \beta \frac{d \log R'(\mathcal{Z}(p, \phi), \phi)}{\log z}} = \frac{\beta}{1 - \beta \frac{R''(\mathcal{Z}(p, \phi), \phi) \mathcal{Z}(p, \phi)}{r(\mathcal{Z}(p, \phi), \phi)}} \quad (76)$$

Note that $r(\mathcal{Z}(p, \phi), \phi)$ is fixed by our choice of projection. Meanwhile

$$R''(\mathcal{Z}(p, \phi), \phi) = \frac{dR'(\mathcal{Z}(p, \phi), \phi)}{dp} \frac{dF(\mathcal{Z}(p, \phi), \phi)}{dz} = \frac{dR'(\mathcal{Z}(p, \phi), \phi)}{dp} f(\mathcal{Z}(p, \phi), \phi) \quad (77)$$

Our projection also fixes $\frac{dR'(\mathcal{Z}(p, \phi), \phi)}{dp}$, so the claim follows from what we have shown for the total income at the p^{th} percentile. $\square$

B.2.2 Non-uniform changes in wages

Proposition. *If Conditions (2) and (3) hold and technical assumptions 3–5 hold, then for all $p \in (0, 1)$,*

$$\frac{\partial \kappa(p, \phi_0)}{\partial \phi_0} < 0 \iff \frac{d}{dp} \mathbb{E} \left[\frac{d \log \omega^h(\phi_0)}{d \phi_0} \mid z^h(\phi_0) = \mathcal{Z}(p, \phi_0) \right] > 0. \quad (78)$$

Proof. We proceed as before to characterize how each term of (57) changes w.r.t ϕ . From Corollary B.2.1, we know for $\phi \in B_\delta(\phi_0)$, the order of households as sorted by their baseline income does not change. An immediate consequence is that how income at the p^{th} percentile changes w.r.t. ϕ is simply how households at the p^{th} percentile of the initial distribution move in response to changing ϕ . Formally,

$$\left. \frac{\partial \mathcal{Z}(p, \phi)}{\partial \phi} \right|_{\phi=\phi_0} = (1 + \beta) \mathcal{Z}(p, \phi_0) \mathbb{E} \left[\left. \frac{d \tilde{w}^h(\phi)}{d \phi} \right| z^h(\phi_0) = \mathcal{Z}(p, \phi_0) \right] \Big|_{\phi=\phi_0} \quad (79)$$

Compared to the neutral benchmark, the impacts of changing ϕ on density is harder to characterize directly. Instead, we will proceed indirectly. By definition,

$$\left. \frac{dF(\mathcal{Z}(p, \phi), \phi)}{d \phi} \right|_{\phi=\phi_0} = f(\mathcal{Z}(p, \phi_0), \phi_0) \frac{\partial \mathcal{Z}(p, \phi)}{\partial \phi} + \left. \frac{\partial F(\mathcal{Z}(p, \phi_0), \phi)}{\partial \phi} \right|_{\phi=\phi_0} = 0 \quad (80)$$

Rearranging the above as a function of z

$$\left. \frac{\partial F(z, \phi)}{\partial \phi} \right|_{\phi=\phi_0} = -f(z, \phi_0) \frac{\partial \mathcal{Z}(F(z, \phi_0), \phi)}{\partial \phi} \Big|_{\phi=\phi_0} \quad (81)$$

Next, if we totally differentiate the density term w.r.t. ϕ ,

$$\left. \frac{df(\mathcal{Z}(p, \phi), \phi)}{d \phi} \right|_{\phi=\phi_0} = f'(\mathcal{Z}(p, \phi_0), \phi_0) \frac{\partial \mathcal{Z}(p, \phi)}{\partial \phi} + \left. \frac{\partial f(\mathcal{Z}(p, \phi_0), \phi)}{\partial \phi} \right|_{\phi=\phi_0} \quad (82)$$

We see the only new component we need to characterize is $\frac{\partial f(z, \phi)}{\partial \phi}$, which we can obtain from differentiating (81).

$$\left. \frac{\partial f(z, \phi)}{\partial \phi} \right|_{\phi=\phi_0} = -f'(z, \phi_0) \frac{\partial \mathcal{Z}(F(z, \phi_0), \phi)}{\partial \phi} - f(z, \phi_0) \frac{d}{dz} \left\{ \frac{\partial \mathcal{Z}(F(z, \phi_0), \phi)}{\partial \phi} \right\} \Big|_{\phi=\phi_0} \quad (83)$$

$$\begin{aligned} &= -f'(z, \phi_0) \frac{\partial \mathcal{Z}(F(z, \phi_0), \phi)}{\partial \phi} \\ &\quad - f(z, \phi_0) (1 + \beta) \frac{d}{dz} \left\{ z \mathbb{E} \left[\left. \frac{d \tilde{w}^h(\phi)}{d \phi} \right| z^h(\phi_0) = z \right] \right\} \Big|_{\phi=\phi_0} \end{aligned} \quad (84)$$

Plugging back in, we have

$$\left. \frac{df(\mathcal{Z}(p, \phi), \phi)}{d\phi} \right|_{\phi=\phi_0} = -f(z, \phi_0)(1 + \beta) \frac{d}{dz} \left\{ z \mathbb{E} \left[\frac{d\tilde{w}^h(\phi)}{d\phi} \middle| z^h(\phi_0) = z \right] \right\} \Big|_{z=\mathcal{Z}(p, \phi_0), \phi=\phi_0} \quad (85)$$

Finally, we wish to characterize $\frac{d\mathbb{E}[\varepsilon^h(\phi)|z^h(\phi)=\mathcal{Z}(p, \phi)]}{d\phi}$. Having conditioned on income and by Condition 3, there is no elasticity heterogeneity. Let $\varepsilon(p, \phi_0) = \mathbb{E}[\varepsilon^h|z^h(\phi_0) = \mathcal{Z}(p, \phi_0)]$.

$$\begin{aligned} \frac{d \log \mathbb{E}[\varepsilon^h|z^h(\phi) = \mathcal{Z}(p, \phi)]}{d\phi} &= \frac{1}{\varepsilon(p, \phi_0)} \frac{d}{d\phi} \frac{\beta}{1 - \beta \frac{\mathcal{Z}(p, \phi) R''(\mathcal{Z}(p, \phi), \phi)}{r(\mathcal{Z}(p, \phi), \phi)}} \\ &= \frac{\varepsilon(p, \phi_0)}{r(z, \phi_0)} \frac{d}{d\phi} \left\{ \mathcal{Z}(p, \phi) R''(\mathcal{Z}(p, \phi), \phi) \right\} \\ &= \frac{\varepsilon(p, \phi_0) R''(\mathcal{Z}(p, \phi_0))}{r(z, \phi_0) f(\mathcal{Z}(p, \phi_0))} \frac{d}{d\phi} \left\{ \mathcal{Z}(p, \phi) f(\mathcal{Z}(p, \phi), \phi) \right\} \\ &= \frac{\varepsilon(p, \phi_0) R''(\mathcal{Z}(p, \phi_0)) \mathcal{Z}(p, \phi_0)}{r(z, \phi_0)} \frac{d}{d\phi} \log \left\{ \mathcal{Z}(p, \phi) f(\mathcal{Z}(p, \phi), \phi) \right\} \\ &= \frac{\varepsilon(p, \phi_0) - \beta}{\beta} \frac{d}{d\phi} \log \left\{ \mathcal{Z}(p, \phi) f(\mathcal{Z}(p, \phi), \phi) \right\} \end{aligned} \quad (86)$$

Altogether,

$$\begin{aligned} \left. \frac{d\tilde{\kappa}(p, \phi)}{d\phi} \right|_{\phi=\phi_0} &= \frac{d \log \{ \mathcal{Z}(p, \phi) f(\mathcal{Z}(p, \phi), \phi) \}}{d\phi} + \frac{d \log \mathbb{E}[\varepsilon^h(\phi)|z^h(\phi) = \mathcal{Z}(p, \phi)]}{d\phi} \Big|_{\phi=\phi_0} \\ &= \frac{\varepsilon(p, \phi_0)}{\beta} \frac{d \log \{ \mathcal{Z}(p, \phi) f(\mathcal{Z}(p, \phi), \phi) \}}{d\phi} \Big|_{\phi=\phi_0} \\ &= -\frac{\varepsilon(p, \phi_0)}{\beta} (1 + \beta) \mathcal{Z}(p, \phi_0) \frac{d}{dz} \left\{ \mathbb{E} \left[\frac{d\tilde{w}^h(\phi)}{d\phi} \middle| z^h(\phi_0) = z \right] \right\} \Big|_{z=\mathcal{Z}(p, \phi_0), \phi=\phi_0} \end{aligned} \quad (87)$$

The claim on the sign immediately follows. $\square$

B.2.3 Elasticity variation across incomes

Proposition. *If Conditions (1) and (3) hold, technical assumptions (3), (4), and (6) hold, and productivities are increasing in ϕ , then for all $p \in (0, 1)$,*

$$\frac{\partial \kappa(p, \phi_0)}{\partial \phi_0} < 0 \quad \Longleftrightarrow \quad \frac{d}{dp} \mathbb{E}[\beta^h \mid z^h(\phi_0) = \mathcal{Z}(p, \phi_0)] > 0 \quad (88)$$

Proof. Again we can invoke Corollary B.2.1 to establish that the order of households does not change for $\phi \in B_\delta(\phi_0)$, so we can proceed in the same fashion as in the previous proposition by characterizing each term of (57). The change in income is immediate,

$$\left. \frac{\partial \mathcal{Z}(p, \phi)}{\partial \phi} \right|_{\phi=\phi_0} = \mathcal{Z}(p, \phi_0)(1 + \beta(\mathcal{Z}(p, \phi_0))) \left. \frac{d\tilde{w}(\phi)}{d\phi} \right|_{\phi=\phi_0} \quad (89)$$

With $\beta(z) = \mathbb{E}[\beta^h \mid z^h(\phi_0) = z]$ as defined previously. Following the same steps to compute the change in density, we have

$$\left. \frac{df(\mathcal{Z}(p, \phi_0), \phi)}{d\phi} \right|_{\phi=\phi_0} = -f(\mathcal{Z}(p, \phi_0), \phi_0) \frac{d\tilde{w}(\phi)}{d\phi} \frac{d}{dz} \left\{ z(1 + \beta(z)) \right\} \Big|_{z=\mathcal{Z}(p, \phi_0), \phi=\phi_0} \quad (90)$$

Lastly, let $\varepsilon(p, \phi_0) = \mathbb{E}[\varepsilon^h \mid z^h(\phi_0) = \mathcal{Z}(p, \phi_0)]$. Having shown the order of households do not change, we have ruled out elasticity sorting, so for $\phi \in B_\delta(\phi_0)$, if $F(z^h(\phi), \phi) = p$, then $\beta^h = \beta(\mathcal{Z}(p, \phi_0))$. By following (86),

$$\begin{aligned} \frac{d \log \mathbb{E}[\varepsilon^h(\phi) \mid z^h(\phi) = \mathcal{Z}(p, \phi)]}{d\phi} &= \frac{1}{\varepsilon(p, \phi_0)} \frac{d}{d\phi} \mathbb{E} \left[\frac{\beta^h}{1 - \beta^h \frac{\mathcal{Z}(p, \phi) R''(\mathcal{Z}(p, \phi), \phi)}{r(\mathcal{Z}(p, \phi), \phi)}} \mid z^h(\phi) = \mathcal{Z}(p, \phi) \right] \\ &= \frac{1}{\varepsilon(p, \phi_0)} \frac{d}{d\phi} \frac{\beta(\mathcal{Z}(p, \phi_0))}{1 - \beta(\mathcal{Z}(p, \phi_0)) \frac{\mathcal{Z}(p, \phi) R''(\mathcal{Z}(p, \phi), \phi)}{r(\mathcal{Z}(p, \phi), \phi)}} \\ &= \frac{\varepsilon(p, \phi_0) - \beta(\mathcal{Z}(p, \phi_0))}{\beta(\mathcal{Z}(p, \phi_0))} \frac{d}{d\phi} \log \left\{ \mathcal{Z}(p, \phi) f(\mathcal{Z}(p, \phi), \phi) \right\} \end{aligned} \quad (91)$$

Putting the pieces together,

$$\begin{aligned} \left. \frac{d\tilde{\kappa}(p, \phi)}{d\phi} \right|_{\phi=\phi_0} &= \frac{d \log \{ \mathcal{Z}(p, \phi) f(\mathcal{Z}(p, \phi), \phi) \}}{d\phi} + \left. \frac{d \log \mathbb{E}[\varepsilon^h(\phi) \mid z^h(\phi) = \mathcal{Z}(p, \phi)]}{d\phi} \right|_{\phi=\phi_0} \\ &= -\frac{\varepsilon(p, \phi_0)}{\beta(\mathcal{Z}(p, \phi_0))} \frac{d\tilde{w}(\phi)}{d\phi} \mathcal{Z}(p, \phi_0) \beta'(\mathcal{Z}(p, \phi_0)) \Big|_{\phi=\phi_0} \end{aligned} \quad (92)$$

The claim on the sign follows immediately. $\square$

B.2.4 Elasticity variation within incomes

Proposition. *Suppose Conditions 1 and 2 hold, Condition 3 fails, the tax schedule $T(\cdot, \phi_0)$ is affine, and productivities increase with ϕ . Then for all $p \in (0, 1)$,*

$$\frac{\partial \kappa(p, \phi_0)}{\partial \phi_0} > 0 \iff \tilde{\alpha}(p, \phi_0) > 0 \quad (93)$$

Proof. Since we are considering an affine tax schedule, keeping the same marginal rate r will trivially satisfy our projection requirement (10). From previous propositions, we see recovering a representation of (81) allows us to characterize how income and density changes at the p^{th} percentile. Recall that the HH FOC is $z^h(\phi) = r^{\beta^h} (w^h(\phi))^{1+\beta^h}$, so the order of households, conditional on β , does not change as we move from ϕ_0 to ϕ given that everyone scales their income by $\left(\frac{w^h(\phi_1)}{w^h(\phi_0)}\right)^{1+\beta} > 0$, which is independent of h under Condition 1. Denote this scaling factor as $c(\phi, \phi_0)^{1+\beta} \equiv \left(\frac{w^h(\phi_1)}{w^h(\phi_0)}\right)^{1+\beta}$, where $c(\phi, \phi_0) > 0$, is differentiable in ϕ , and $c(\phi_0, \phi_0) = 1$. By conditioning on a particular value of β ,

$$F(z|\beta, \phi) = F\left(\frac{z}{\left(\frac{w^h(\phi)}{w^h(\phi_0)}\right)^{1+\beta}} \middle| \beta, \phi_0\right) \implies \frac{\partial F(z|\beta, \phi)}{\partial \phi} \bigg|_{\phi=\phi_0} = -f(z|\beta, \phi_0) z (1+\beta) \frac{dc(\phi, \phi_0)}{d\phi} \bigg|_{\phi=\phi_0} \quad (94)$$

Plugging the above equation and define $\gamma \equiv \frac{dc(\phi, \phi_0)}{d\phi} \bigg|_{\phi=\phi_0}$,

$$\begin{aligned} \frac{\partial F(z, \phi)}{\partial \phi} \bigg|_{\phi=\phi_0} &= \int \frac{\partial F(z|\beta^h, \phi)}{\partial \phi} \bigg|_{\phi=\phi_0} dF(\beta^h) \quad (\text{the marginal over } \beta \text{ is independent of } \phi) \\ &= -\gamma z \int f(z|\beta^h, \phi_0) (1 + \beta^h) dF(\beta^h) \\ &= -\gamma z f(z, \phi_0) (1 + \mathbb{E}[\beta^h | z^h(\phi_0) = z]) = -\gamma z f(z, \phi_0) (1 + \mathbb{E}[\beta^h]) \end{aligned}$$

The last line uses Assumption (2), which stipulates that the within income distribution of β^h is the same across initial incomes, so in particular the means are identical. From (81), we immediately have

$$\frac{\partial \mathcal{Z}(p, \phi)}{\partial \phi} \bigg|_{\phi=\phi_0} = \gamma \mathcal{Z}(p, \phi_0) (1 + \mathbb{E}[\beta^h]) \quad (95)$$

$$\frac{\partial f(\mathcal{Z}(p, \phi_0), \phi)}{\partial \phi} \bigg|_{\phi=\phi_0} = -\gamma \mathcal{Z}(p, \phi_0) f'(\mathcal{Z}(p, \phi_0), \phi_0) - \gamma f(\mathcal{Z}(p, \phi_0), \phi_0) (1 + \mathbb{E}[\beta^h]) \quad (96)$$

Combining the two by (82),

$$\frac{df(\mathcal{Z}(p, \phi), \phi)}{d\phi} \bigg|_{\phi=\phi_0} = -\gamma f(\mathcal{Z}(p, \phi_0), \phi_0) (1 + \mathbb{E}[\beta^h]) \quad (97)$$

We now move to characterizing $\frac{d\mathbb{E}[\varepsilon^h(\phi)|z^h(\phi)=\mathcal{Z}(p,\phi)]}{d\phi}$. Note with a linear tax $\varepsilon^h = \beta^h$.

$$\begin{aligned}
\left. \frac{d\mathbb{E}[\varepsilon^h(\phi)|z^h(\phi) = \mathcal{Z}(p, \phi)]}{d\phi} \right|_{\phi=\phi_0} &= \left. \frac{d}{d\phi} \int \beta^h f(\beta^h|\mathcal{Z}(p, \phi), \phi) d\beta^h \right|_{\phi=\phi_0} \\
&= \left. \frac{d}{d\phi} \frac{\int \beta^h f(\mathcal{Z}(p, \phi)|\beta^h, \phi) f(\beta^h) d\beta^h}{f(\mathcal{Z}(p, \phi), \phi)} \right|_{\phi=\phi_0} \\
&= \frac{\int \beta^h \frac{df(\mathcal{Z}(p, \phi)|\beta^h, \phi)}{d\phi} \Big|_{\phi=\phi_0} f(\beta^h) d\beta^h}{f(\mathcal{Z}(p, \phi_0), \phi_0)} \\
&\quad - \underbrace{\frac{\frac{df(\mathcal{Z}(p, \phi), \phi)}{d\phi} \Big|_{\phi=\phi_0} \int \beta^h f(\mathcal{Z}(p, \phi_0)|\beta^h, \phi_0) f(\beta^h) d\beta^h}{f(\mathcal{Z}(p, \phi_0), \phi_0)^2}}_{-\gamma(1+\mathbb{E}[\beta^h])\mathbb{E}[\beta^h]}
\end{aligned} \tag{98}$$

The only unknown component in (98) is $\frac{df(\mathcal{Z}(p, \phi)|\beta, \phi)}{d\phi}$, the density of income at the p^{th} percentile in the β -conditional distribution. We can compute it directly using (94)

$$\begin{aligned}
\left. \frac{df(\mathcal{Z}(p, \phi)|\beta, \phi)}{d\phi} \right|_{\phi=\phi_0} &= f'(\mathcal{Z}(p, \phi_0)|\beta) \frac{\partial \mathcal{Z}(p, \phi)}{\partial \phi} + \left. \frac{\partial f(\mathcal{Z}(p, \phi_0)|\beta, \phi)}{\partial \phi} \right|_{\phi=\phi_0} \\
&= \mathcal{Z}(p, \phi_0) f'(\mathcal{Z}(p, \phi_0)|\beta, \phi) \gamma(\mathbb{E}[\beta^h] - \beta) - \gamma(1 + \beta) f(\mathcal{Z}(p, \phi_0)|\beta, \phi) \Big|_{\phi=\phi_0} \\
&= \mathcal{Z}(p, \phi_0) f'(\mathcal{Z}(p, \phi_0), \phi) \frac{f(\beta|\mathcal{Z}(p, \phi_0), \phi)}{f(\beta)} \gamma(\mathbb{E}[\beta^h] - \beta) \\
&\quad - \gamma(1 + \beta) f(\mathcal{Z}(p, \phi_0)|\beta, \phi) \Big|_{\phi=\phi_0}
\end{aligned} \tag{99}$$

Where the last line again follows from Assumption (2). Plugging (99) into (98) gives

$$\begin{aligned}
\left. \frac{d\mathbb{E}[\varepsilon^h(\phi)|z^h(\phi) = \mathcal{Z}(p, \phi)]}{d\phi} \right|_{\phi=\phi_0} &= \frac{\int \beta^h \frac{df(\mathcal{Z}(p, \phi)|\beta^h, \phi)}{d\phi} \Big|_{\phi=\phi_0} f(\beta^h) d\beta^h}{f(\mathcal{Z}(p, \phi), \phi)} + \gamma(1 + \mathbb{E}[\beta^h]) \mathbb{E}[\beta^h] \Big|_{\phi=\phi_0} \\
&= -\gamma \mathbb{V}[\beta^h|z^h(\phi_0) = \mathcal{Z}(p, \phi_0)] \frac{\mathcal{Z}(p, \phi_0) f'(\mathcal{Z}(p, \phi_0), \phi_0)}{f(\mathcal{Z}(p, \phi_0), \phi_0)} \\
&\quad - \gamma(\mathbb{E}[\beta^h] + \mathbb{E}[(\beta^h)^2]) + \gamma(1 + \mathbb{E}[\beta^h]) \mathbb{E}[\beta^h] \Big|_{\phi=\phi_0} \\
&= -\gamma \mathbb{V}[\beta^h|z^h(\phi_0) = \mathcal{Z}(p, \phi_0)] \left(1 + \frac{\mathcal{Z}(p, \phi_0) f'(\mathcal{Z}(p, \phi_0), \phi_0)}{f(\mathcal{Z}(p, \phi_0), \phi_0)} \right) \Big|_{\phi=\phi_0}
\end{aligned} \tag{100}$$

Putting the pieces together,

$$\begin{aligned}
\left. \frac{d\tilde{\kappa}(p, \phi)}{d\phi} \right|_{\phi=\phi_0} &= \frac{d \log\{\mathcal{Z}(p, \phi) f(\mathcal{Z}(p, \phi), \phi)\}}{d\phi} + \left. \frac{d \log \mathbb{E}[\varepsilon^h(\phi)|z^h(\phi) = \mathcal{Z}(p, \phi)]}{d\phi} \right|_{\phi=\phi_0} \\
&= -\gamma \frac{\mathbb{V}[\beta^h|z^h(\phi_0) = \mathcal{Z}(p, \phi_0)]}{\mathbb{E}[\beta^h] f(\mathcal{Z}(p, \phi_0), \phi_0)} \frac{d\{zf(z, \phi_0)\}}{dz} \Big|_{z=\mathcal{Z}(p, \phi_0), \phi=\phi_0}
\end{aligned} \tag{101}$$

We remind that

$$\alpha(p, \phi) = - \frac{d \log (zf(z, \phi))}{d \log z} \Big|_{z=\mathcal{Z}(p, \phi)} \quad (102)$$

So the claim follow immediately. $\square$

C Empirical Appendix

C.1 Tax Reforms According to TAXSIM

Here we probe the source of our identifying variation, tax reforms across states and over time. In our sample, 5.7% of single filers and 5.5% of the married filers experienced a three-year ahead simulated change in marginal tax rate exceeding 5% in magnitude. Figure 6 plots the population-weighted counts of filers experiencing a substantial tax change by state.

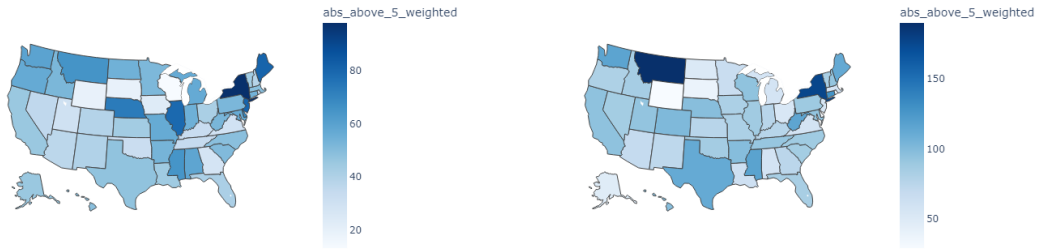


Figure 6: Observations experiencing a tax reform exceeding 5% in marginal tax rate by state. Left panel: single filers. Right panel: joint filers. Counts are weighted by population.

C.2 Constant Taxable Income Definition

To avoid artificial fluctuations in taxable income from one period to the next introduced by changing definitions of taxable income, we adopt a definition of taxable income—in line with the definition of “broad income” in [Gruber and Saez \[2002\]](#)—based on variables available to us in SIPP. Using TAXSIM variable names, our taxable income is defined by

$$\begin{aligned} \text{Taxable Income} \equiv & \text{pwages}_r + \text{swages}_r + \text{dividends}_r + \text{intrec}_r \\ & + \text{pensions}_r + \text{pui}_r + \text{sui}_r + \text{nonprop}_r \\ & + \text{otherprop}_r + \text{pbusinc}_r + \text{sbusinc}_r \end{aligned}$$

The subscript r denotes that the variables are in real terms, unlike inputs into TAXSIM. To adjust for inflation, we divide all monetary variables by the PCE of that month as published by the Federal Reserve before proceeding with our usual aggregating proce-

dure.¹⁹

C.3 Applying TAXSIM to SIPP

We use all waves of the 2001, 2004 and 2008 Survey of Income and Program Participation (SIPP) core panels, covering years 2000 to 2013. Since observations are at an individual-year-month level, where applicable, we adjust these monthly observations to an yearly level. Let $x_{i,j,y}$ be the observation for some individual i at month j of year y . Define

$$\bar{x}_{i,y} \equiv \left(\sum_{j \in M_{i,x,y}} x_{i,j,y} \right) \cdot \frac{12}{|M_{i,x,y}|} \quad (103)$$

where $M_{i,x,y}$ represent the subset of months we have observations for variable x of individual i during year y .

Table 1 maps SIPP variables to TAXSIM 35 variables. For notational simplicity, this aggregation procedure is denoted with $\dagger$ in the table; $(\cdot)^\dagger$ means first computing $(\cdot)$ at a monthly level before applying the aggregation procedure. Note that certain inputs labeled as “not available in core panel” may be available in a subset of the four SIPP panels we studied, but since we are looking for a consistent definition across the board, such inputs are omitted.

Several variables in TAXSIM 35, while not directly present in the SIPP panels, can be obtained if we link couples using available variables. The combination of sample unit ID (SSUID), person number of the reference person (EFREFPER), and person number of the reference person’s spouse (EFSPOUSE) uniquely identifies a couple. This gives us SAGE, SWAGES, SUI, and SBUSINC.

After data pre-processing, we compute state and federal marginal rates via TAXSIM 35. Let $x_i \in \mathbb{R}^n$ be a vector of TAXSIM 35 inputs associated with individual i . Note that one of the variables in x_i is year. Take $y_{0,i}$ to mean the actual year this individual-year observation was recorded and let $x_{y,i}$ denote setting the year variable to y while holding everything else in the vector unchanged. Additionally, take $\text{adj}(x_{y,i})$ to mean multiplying each of the monetary variables in x_i by $\text{PCE}_y/\text{PCE}_{y_{0,i}}$. Lastly, take $\text{TAXSIM 35}(\cdot)$ to mean a function that takes vectors of the form x_i and returns $(\text{frate}_i, \text{srate}_i)$, which are the federal and state marginal rate for individual i as predicted by TAXSIM 35.

With these formalisms introduced, our procedure boils down to the following: for each $j \in \{0, 1, 2, 3\}$, we collect

$$\text{TAXSIM 35}(\text{adj}(x_{y_{0,i}+j}))$$

We update the year variable up to 3 years ahead, adjust monetary variables for inflation using PCE, and apply TAXSIM 35 to compute the marginal rate of the same individual had she kept the same earnings profile in a world governed by tax laws j years ahead.

¹⁹<https://fred.stlouisfed.org/series/PCEPI>

Table 1: **Conversion between SIPP and TAXSIM 35 variables**

TAXSIM 35	SIPP
taxsimid	Not relevant
year	RHCALYR
state	TFIPSST
mstat	EMS ²⁰
page	TAGE
sage	From linking couples
depx	RFNKIDS
age1	Not available in core panels
age2	Not available in core panels
age3	Not available in core panels
pwages	TPEARNT [†]
swages	From linking couples
psemp	Not available in core panels
ssemp	Not available in core panels
dividends	TDIVINC [†]
intrec	TINTINC [†]
stcg	Not available in core panels
ltcg	Not available in core panels
otherprop	(TOACLR + TJACLR) [†]
nonprop	(T29AMT + T36AMT + TRNDUP1 + TRNDUP2) [†]
pensions	(T30AMT + T31AMT + T32AMT + T34AMT + T35AMT + T39AMT + T42AMT) [†]
gssi	(T01AMTA + T01AMTK) [†]
pui	T05AMT [†]
sui	From linking couples
transfers	TPTRNINC
rentpaid	TMTHRNT [†]
proptax	Not available in core panels
otheritem	Not available in core panels
childcare	Not available in core panels
mortgage	TMOWN [†]
scorp	Not available in core panels
pbusinc	(TBMSUM1 + TPMUSUM2) [†]
pprofin	Not available in core panels
sbusinc	From linking couples
sprofin	Not available in core panels

C.4 GRF

C.4.1 Details of GRF elasticity estimation

When applying instrumental forests, we do not center our estimates using the default option that uses random forests. Instead, we estimate the following regression

$$\Delta \log z_{i,t+3} = \alpha + \delta_{s,y} + \gamma M_{i,t} + \sum_{j=1}^{10} \theta_j \text{SPLINE}_j(y_{i,t}) + \epsilon_{i,t} \quad (104)$$

Where $\Delta \log z_{i,t+3}$ is the three-year change in log taxable income, $\delta_{s,y}$ is the state-year fixed effect, $M_{i,t}$ is a marital status dummy, and $\{\text{SPLINE}_j\}$ are the 10-piece linear splines. We apply estimates of this regression in the “Y.hat” option of the R package for instrumental forests. We repeat this step for “Z.hat” and “W.hat” with the same estimating equation but replacing the LHS as the instrument and the treatment, respectively.

Finally, our set of predictive variables are the following. The demographic variables are sex, marital status, number of dependents, education, and age. The job related variables are years at current job and base year occupation category. Other tax related variables include base year taxable income, base year marginal rate, base year self-employment income, base year mortgage interest, base year rent, and base year property tax.

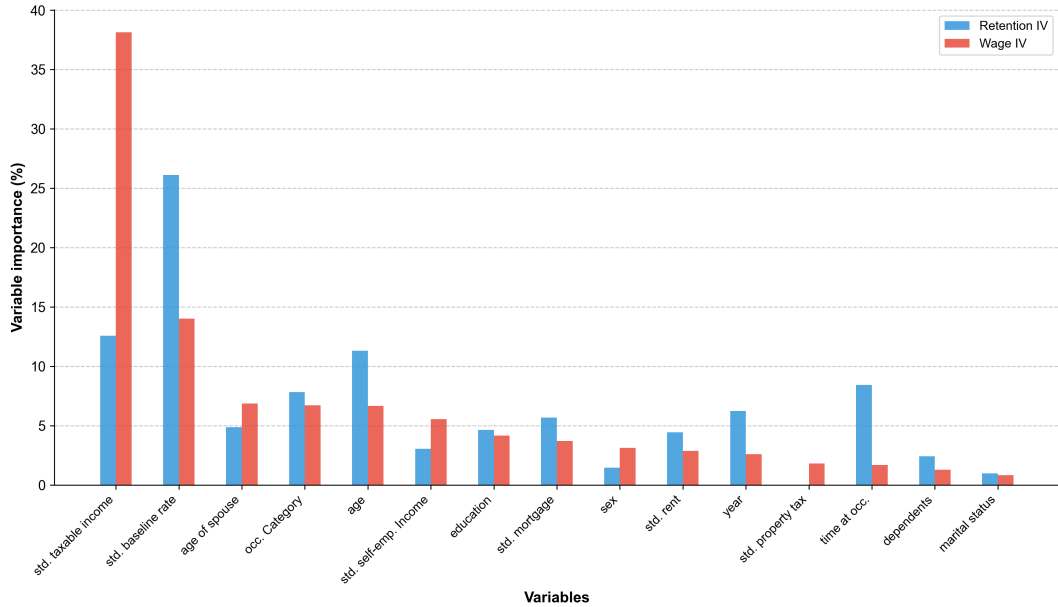


Figure 7: Variable importance measure from GRF estimation.

Our procedure for estimating the wage elasticity is identical except for using a different treatment and a different instrument. Figure 7 plots the “variable importance” measure provided by the instrumental forests package. These estimates reflect how often the algorithm split along a particular variable when growing trees.

C.5 Backing Out Income-Conditional Correlation

Let β^h and ε_w^h denote the true individual elasticities. We would like to know the average income-conditional correlation of elasticities of taxable income out of tax and wage changes. Formally, our statistic of interest is

$$\psi(z) \equiv \text{Corr}[\beta^h, \varepsilon_w^h | z^h = z] = \frac{\text{Cov}[\beta^h, \varepsilon_w^h | z^h = z]}{\text{SD}[\beta^h | z^h = z] \text{SD}[\varepsilon_w^h | z^h = z]} \quad (105)$$

To make progress, we maintain the following assumption:

Assumption 7. *Conditional on the vector of controls x used in GRF estimation, there is no within-income variation in elasticities. Formally,*

$$\text{Var}(\varepsilon_w^h | z, x) = \text{Var}(\beta^h | z, x) = 0 \quad (106)$$

Let $\hat{\beta}_{i,t}$ and $\hat{\varepsilon}_{w,i,t}$ denote the individual estimates from GRF. Under Assumption 7, the laws of total variance and covariance imply

$$\psi(z) = \mathbb{E} \left[\frac{\text{Cov}[\hat{\beta}_{i,t}, \hat{\varepsilon}_{w,i,t} | z^h = z]}{\text{SD}[\hat{\beta}_{i,t} | z^h = z] \text{SD}[\hat{\varepsilon}_{w,i,t} | z^h = z]} \right] \quad (107)$$

where we have used the observation that two variables with no (conditional) variance cannot (conditionally) covary. Note that the expectation is taken over draws of the data.

C.5.1 Estimation procedure

In practice, instead of using log income as our running variable, we use year-demeaned log incomes $\tilde{z}$ to filter out business cycle variation in the income distribution.

1. Form GRF estimates $\hat{\beta}_{i,t}$ and $\hat{\varepsilon}_{w,i,t}$.
2. Form estimates of conditional average tax and wage elasticities as functions of year-demeaned log-income by running the following local linear regressions²¹

$$\begin{aligned} \hat{\beta}_{i,t} &= f_{\beta}(\tilde{z}_{i,t}) + \epsilon_{i,t}^{\beta} \\ \hat{\varepsilon}_{w,i,t} &= f_{\varepsilon_w}(\tilde{z}_{i,t}) + \epsilon_{i,t}^{\varepsilon_w} \end{aligned} \quad (108)$$

3. Form estimates of conditional covariances and variances of tax and wage elasticities as functions of year-demeaned log income by running the following local linear

²¹For all local regressions, we use a bandwidth of 0.2. Eventually we will apply a standard data-driven procedure for bandwidth selection.

regressions

$$\begin{aligned}
(\hat{\beta}_{i,t} - \hat{f}_\beta(\tilde{z}_{i,t})) (\hat{\epsilon}_{w,i,t} - \hat{f}_{\epsilon_w}(\tilde{z}_{i,t})) &= \hat{f}_c(\tilde{z}_{i,t}) + \epsilon_{i,t}^c \\
(\hat{\beta}_{i,t} - \hat{f}_\beta(\tilde{z}_{i,t}))^2 &= \hat{f}_{v_\beta}(\tilde{z}_{i,t}) + \epsilon_{i,t}^{v_\beta} \\
(\hat{\epsilon}_{w,i,t} - \hat{f}_{\epsilon_w}(\tilde{z}_{i,t}))^2 &= \hat{f}_{v_{\epsilon_w}}(\tilde{z}_{i,t}) + \epsilon_{i,t}^{v_{\epsilon_w}}
\end{aligned} \tag{109}$$

4. Form a plug-in estimate $\hat{\psi}(z)$ of ψ as follows:

$$\hat{\psi}(\tilde{z}) = \hat{f}_c(\tilde{z}) / \left(\sqrt{\hat{f}_{v_\beta}(\tilde{z})} \sqrt{\hat{f}_{v_{\epsilon_w}}(\tilde{z})} \right) \tag{110}$$

Currently, we plot $\hat{\psi}(z)$ on a grid $z \in \mathcal{Z}$ defined as the percentiles of the income distribution (this is the income distribution in the data; we use this same income grid within all bootstraps).

5. For each of $B = 300$ bootstrap samples b , generate a bootstrap sample of the dataset and repeat steps 1–4 to obtain a bootstrap estimate $\hat{\psi}_b$.²²
6. Compute the bootstrap-bias-corrected estimate $\hat{\psi}_{BC}$ and $\hat{\psi}_{BC}(z \in \mathcal{Z})$ and 95% confidence interval $[\hat{\psi}_L, \hat{\psi}_H]$ and $[\hat{\psi}_L(z \in \mathcal{Z}), \hat{\psi}_H(z \in \mathcal{Z})]$ as follows:

$$\begin{aligned}
\hat{\psi}_{BC} &\equiv 2\hat{\psi} - \text{mean}(\{\hat{\psi}_b\}_{b=1}^B) \\
\hat{\psi}_L &\equiv 2\hat{\psi} - \text{percentile}_{97.5}(\{\hat{\psi}_b\}_{b=1}^B) \\
\hat{\psi}_H &\equiv 2\hat{\psi} - \text{percentile}_{2.5}(\{\hat{\psi}_b\}_{b=1}^B) \\
\hat{\psi}_{BC}(z) &\equiv 2\hat{\psi}(z) - \text{mean}(\{\hat{\psi}_b(z)\}_{b=1}^B) \\
\hat{\psi}_L(z) &\equiv 2\hat{\psi}(z) - \text{percentile}_{97.5}(\{\hat{\psi}_b(z)\}_{b=1}^B) \\
\hat{\psi}_H(z) &\equiv 2\hat{\psi}(z) - \text{percentile}_{2.5}(\{\hat{\psi}_b(z)\}_{b=1}^B)
\end{aligned}$$

²²We will eventually set $B = 1000$.